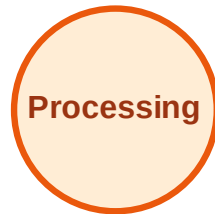


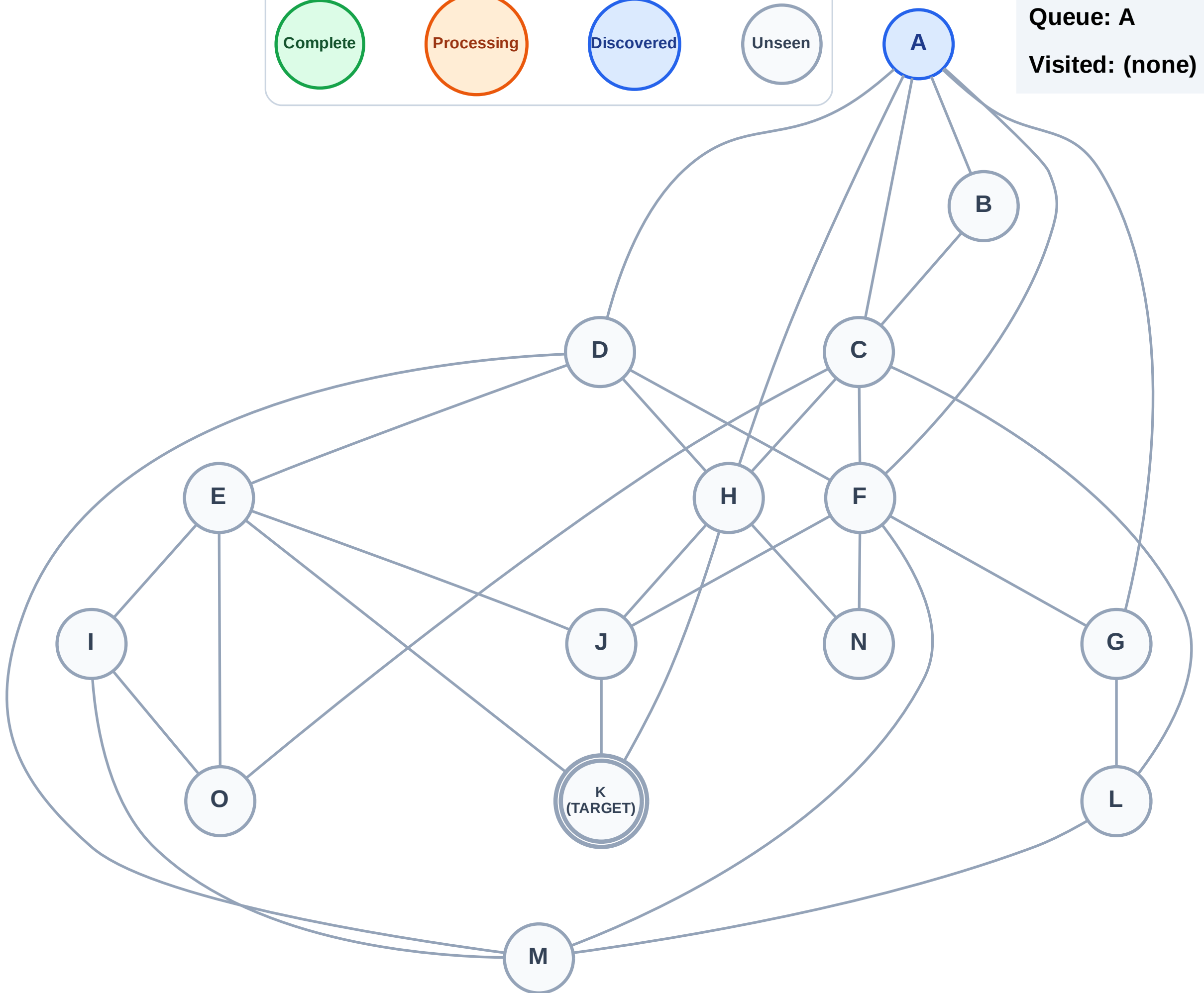
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Queue: A
Visited: (none)



BFS Traversal

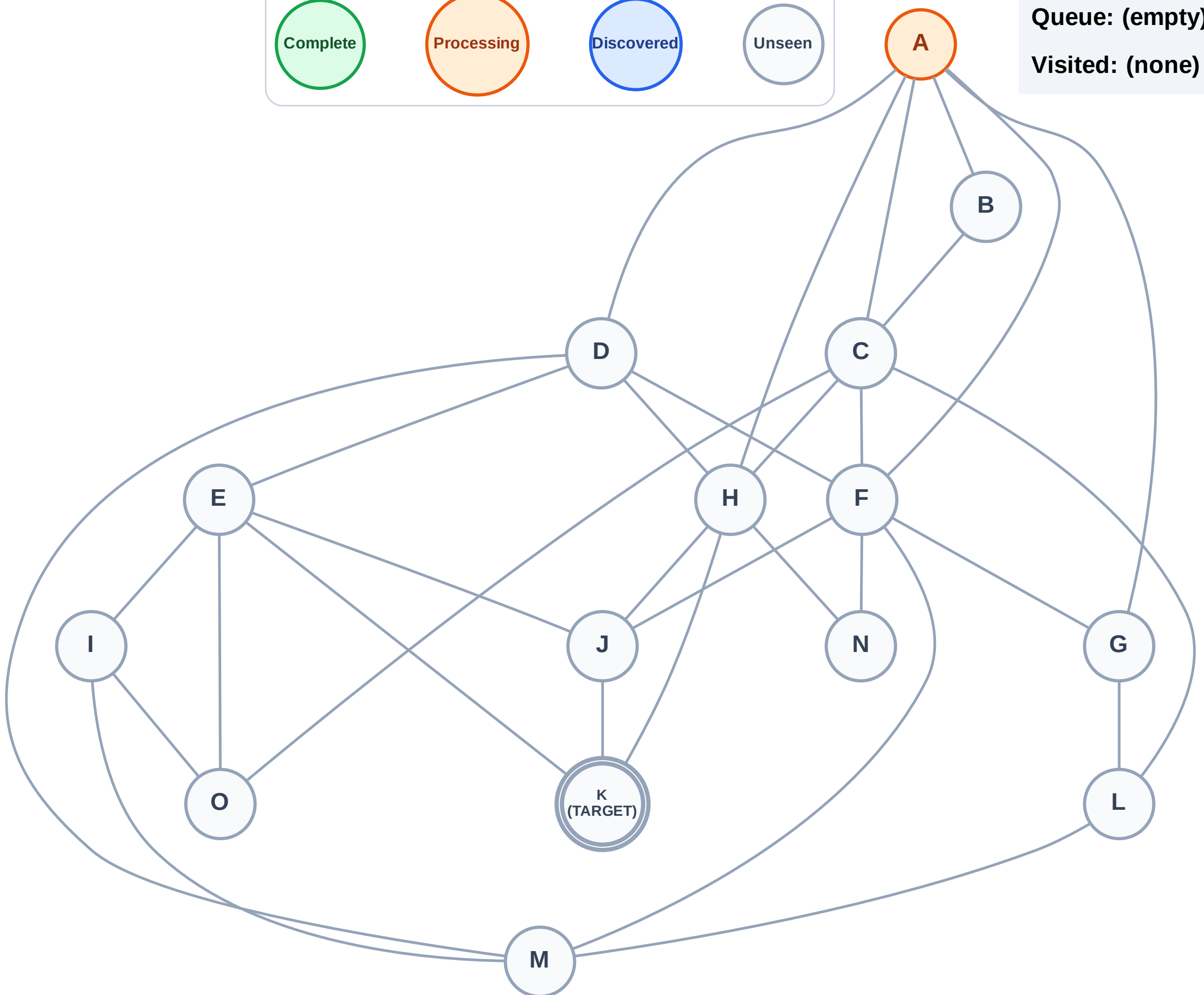
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



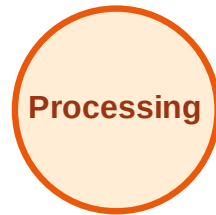
BFS Traversal

Step 3: Discover B, C, D, F, G, H from A

Legend



Complete



Processing



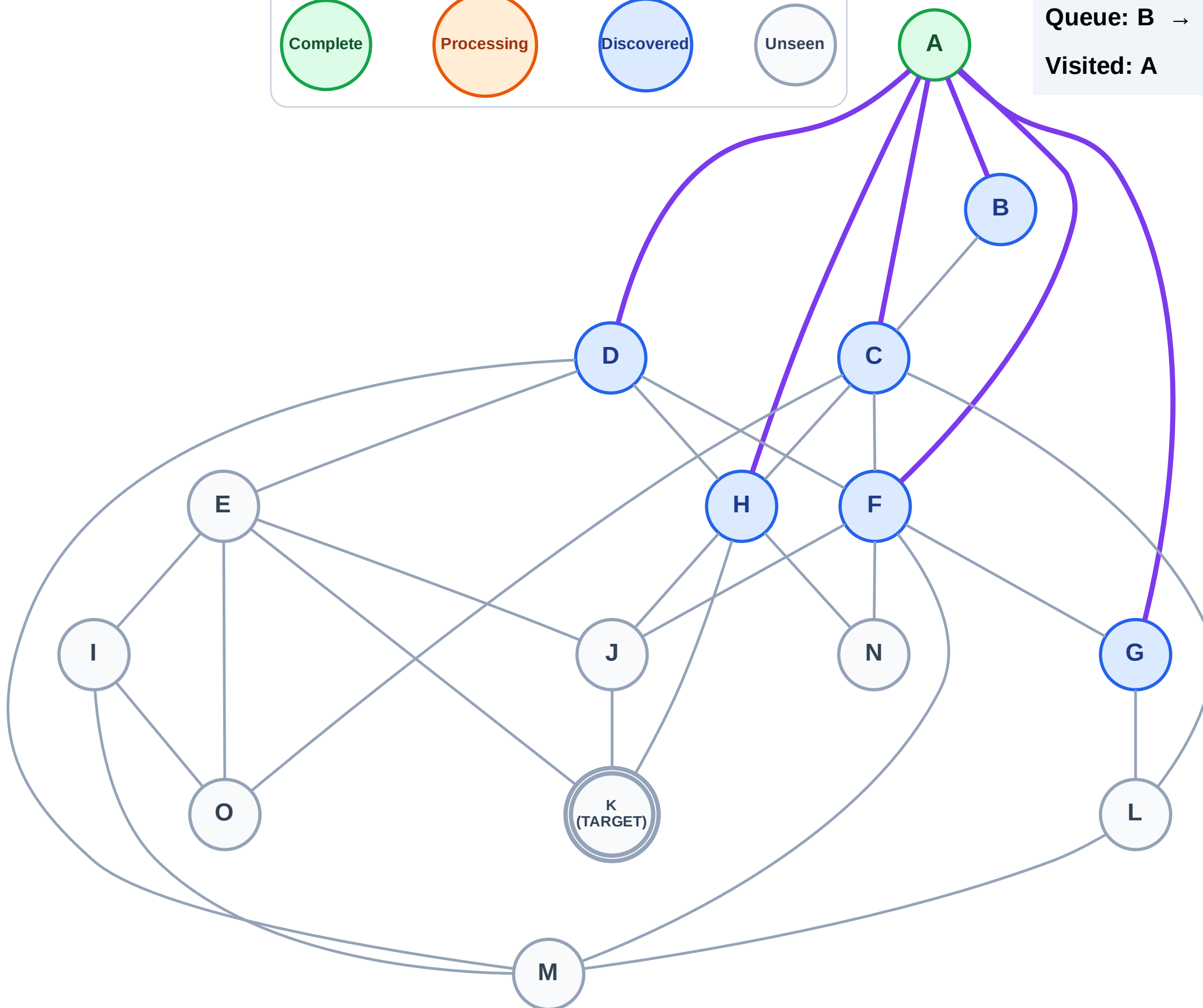
Discovered



Unseen

Queue: B → C → D → F → G → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

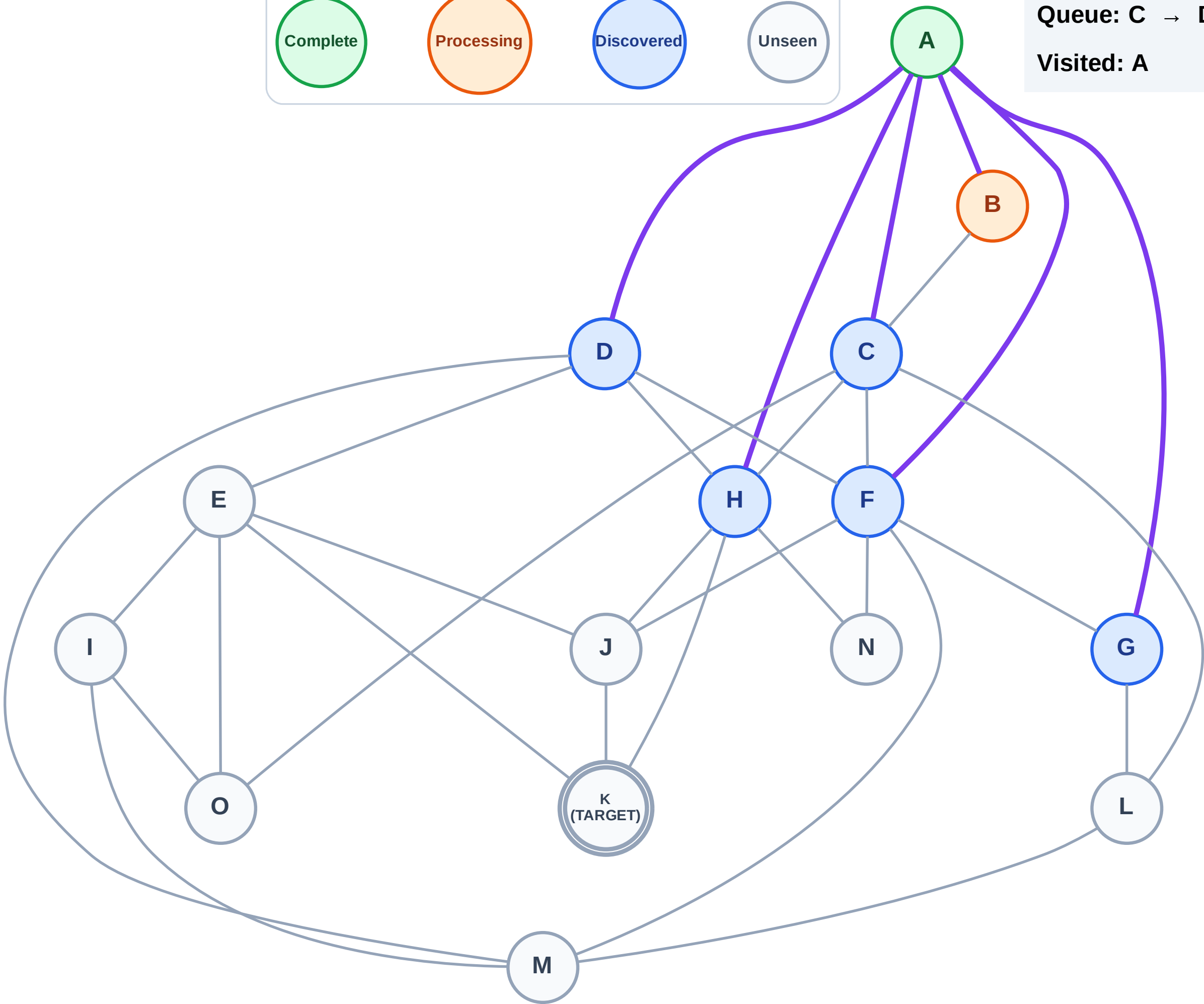
Processing

Discovered

Unseen

Queue: C → D → F → G → H

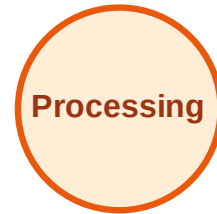
Visited: A



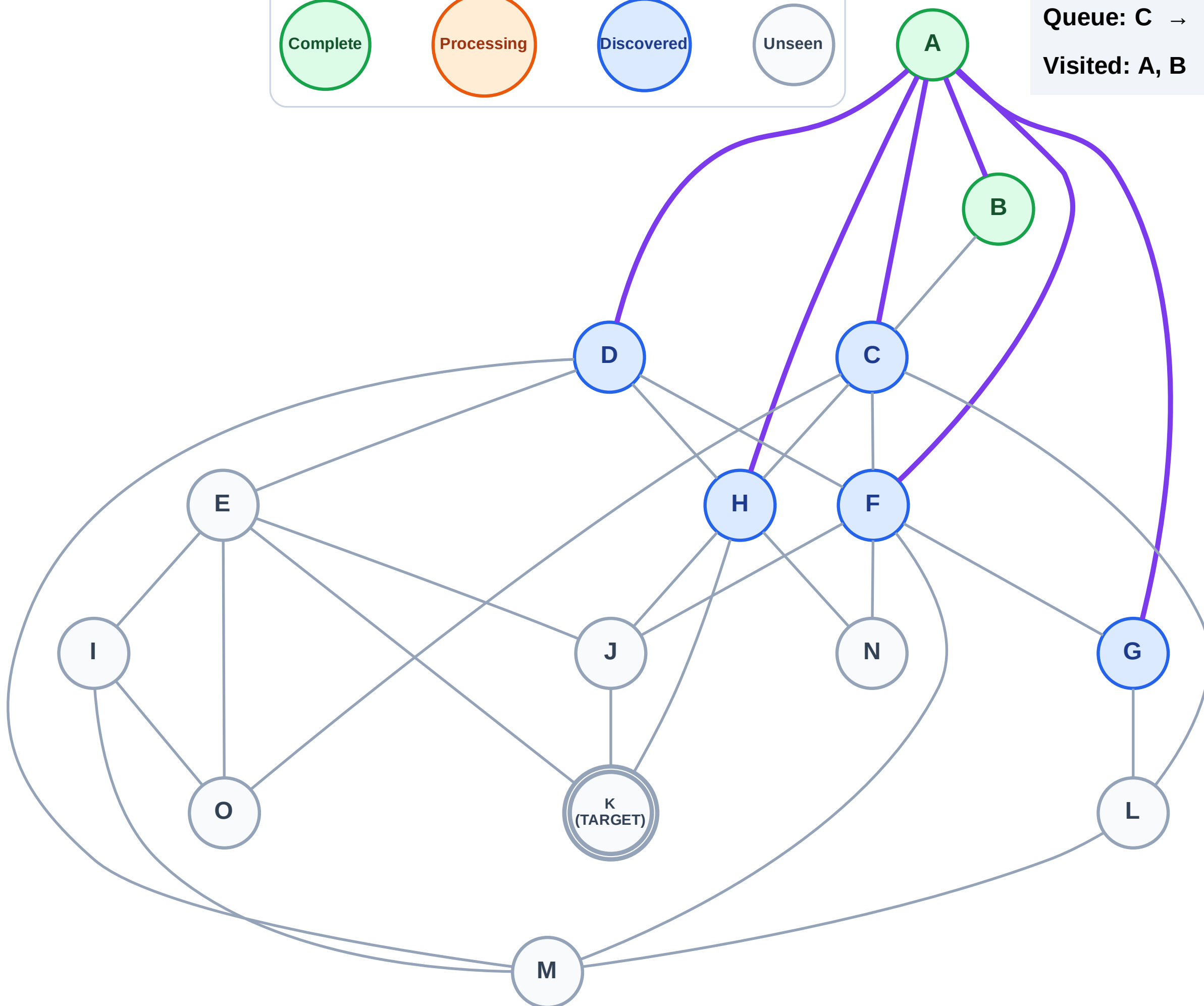
BFS Traversal

Step 5: No new neighbors from B

Legend



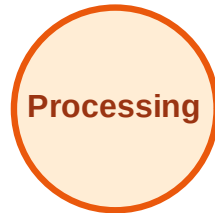
Queue: C → D → F → G → H
Visited: A, B



BFS Traversal

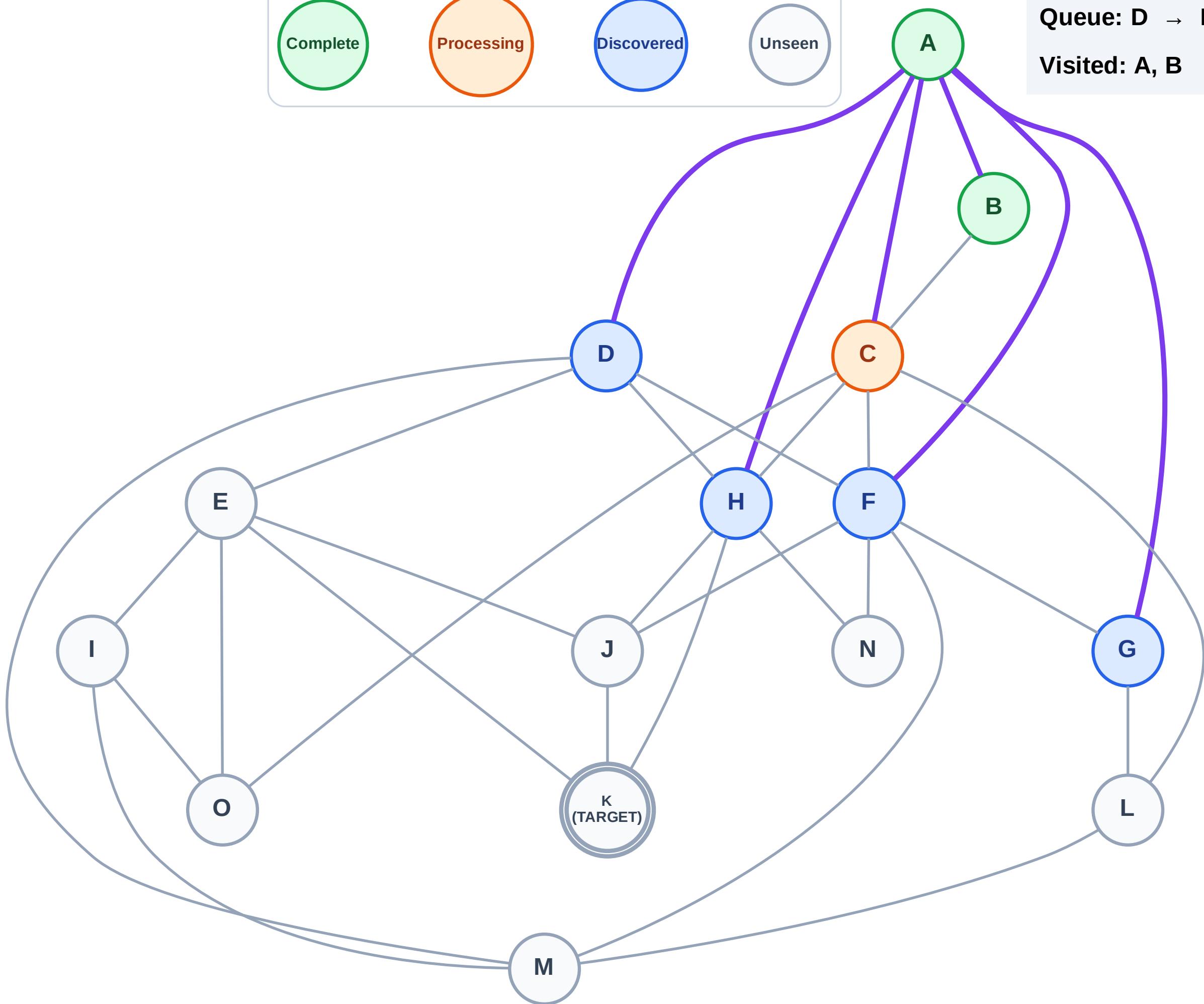
Step 6: Process node C

Legend



Queue: D → F → G → H

Visited: A, B



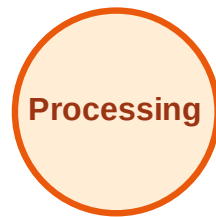
BFS Traversal

Step 7: Discover L, O from C

Legend



Complete



Processing



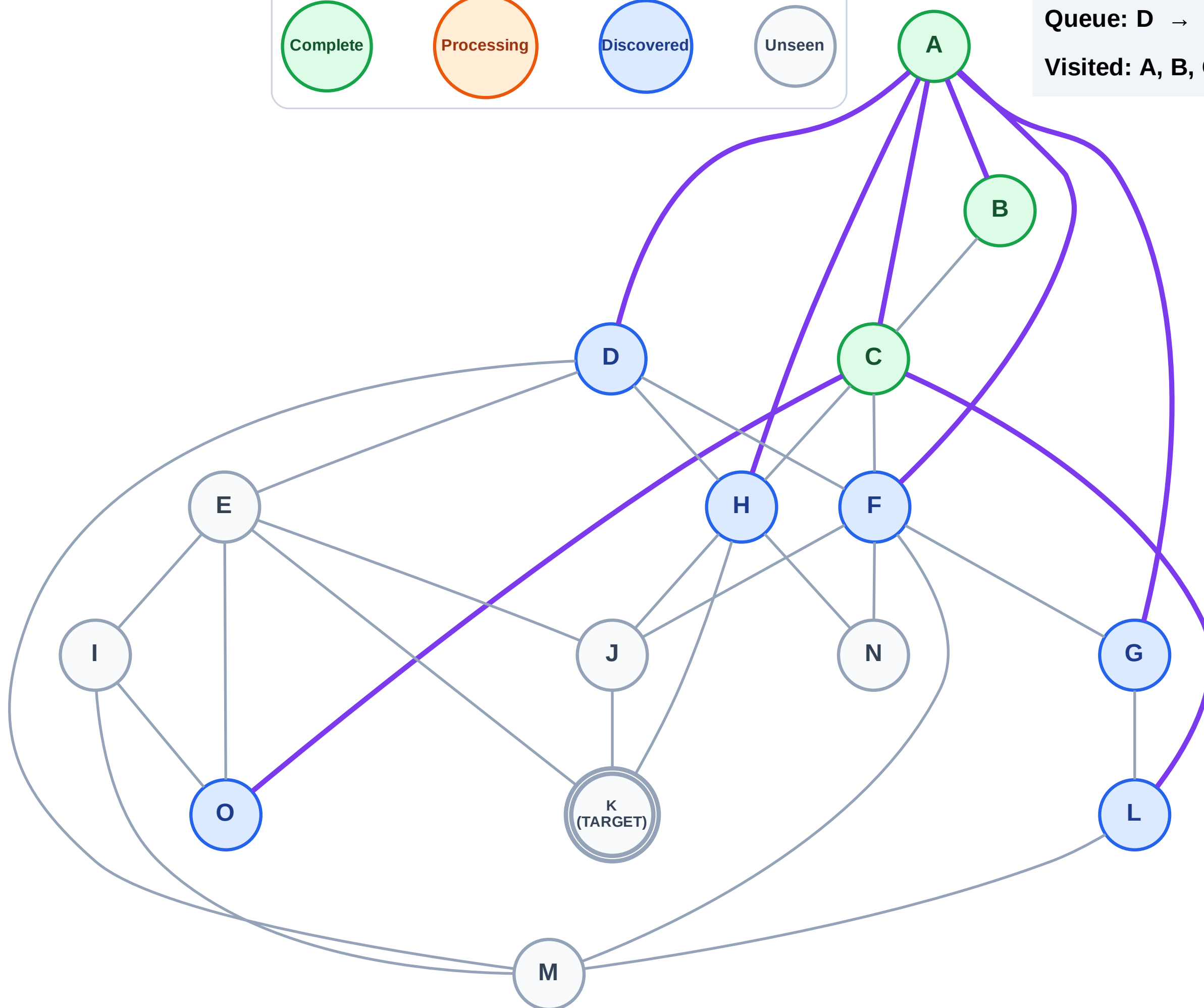
Discovered



Unseen

Queue: D → F → G → H → L → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

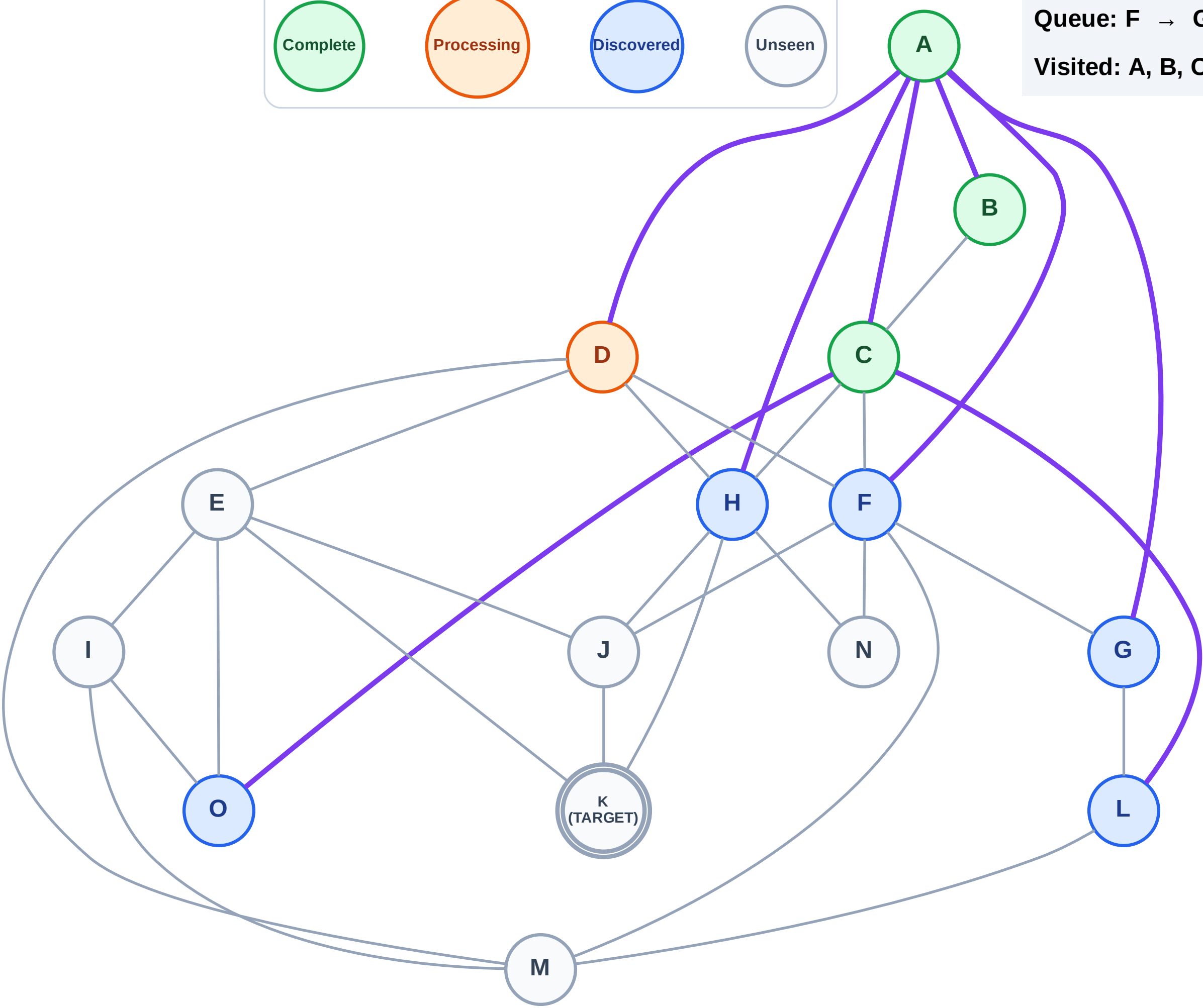
Processing

Discovered

Unseen

Queue: F → G → H → L → O

Visited: A, B, C



Step 9: Discover E, M from D

Legend



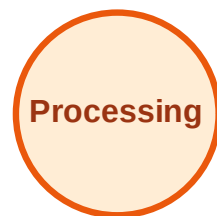
BFS Traversal

Step 10: Process node F

Legend



Complete



Processing



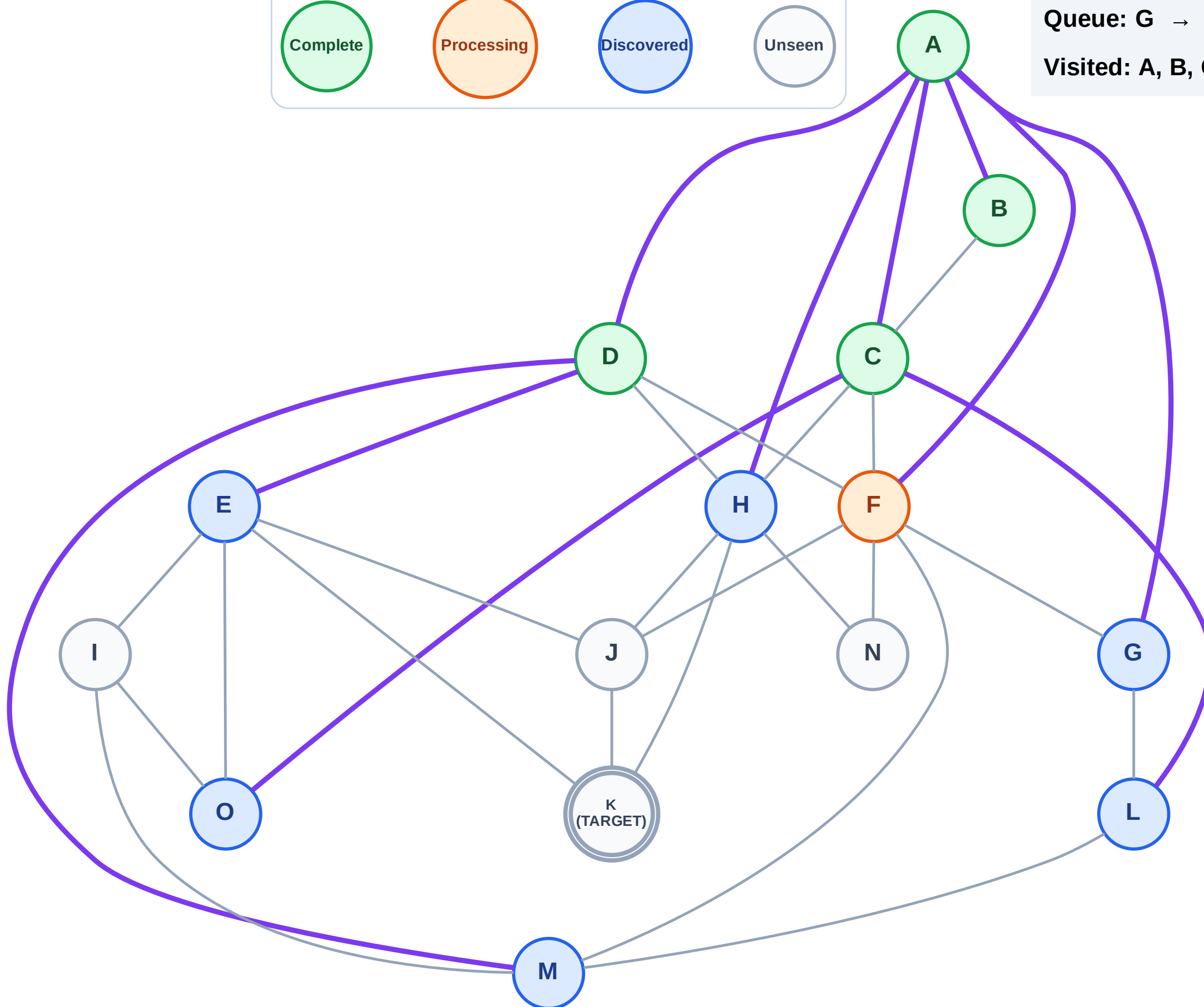
Discovered



Unseen

Queue: G → H → L → O → E → M

Visited: A, B, C, D



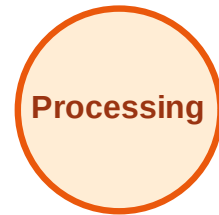
BFS Traversal

Step 11: Discover J, N from F

Legend



Complete



Processing



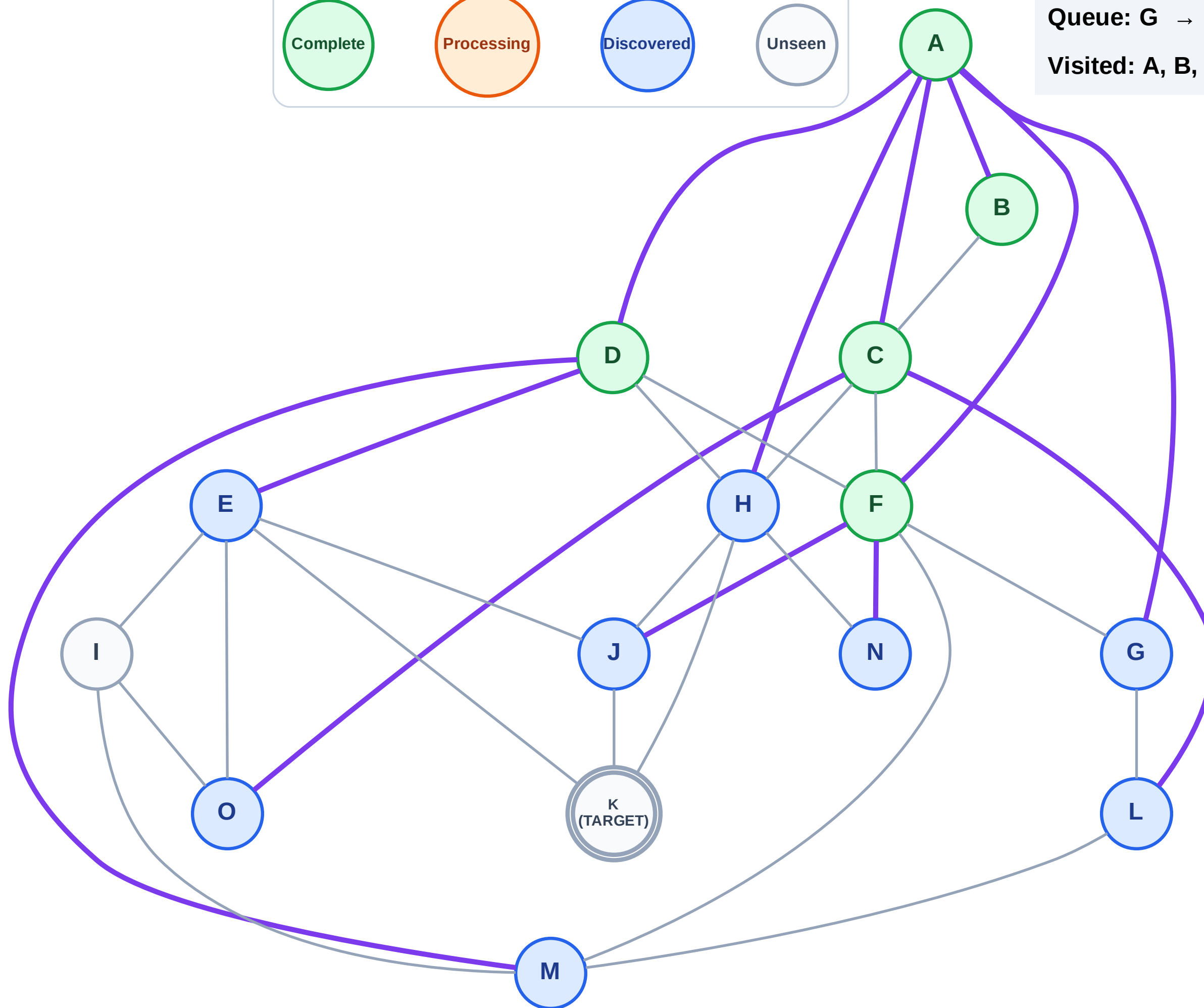
Discovered



Unseen

Queue: G → H → L → O → E → M → J → N

Visited: A, B, C, D, F



BFS Traversal

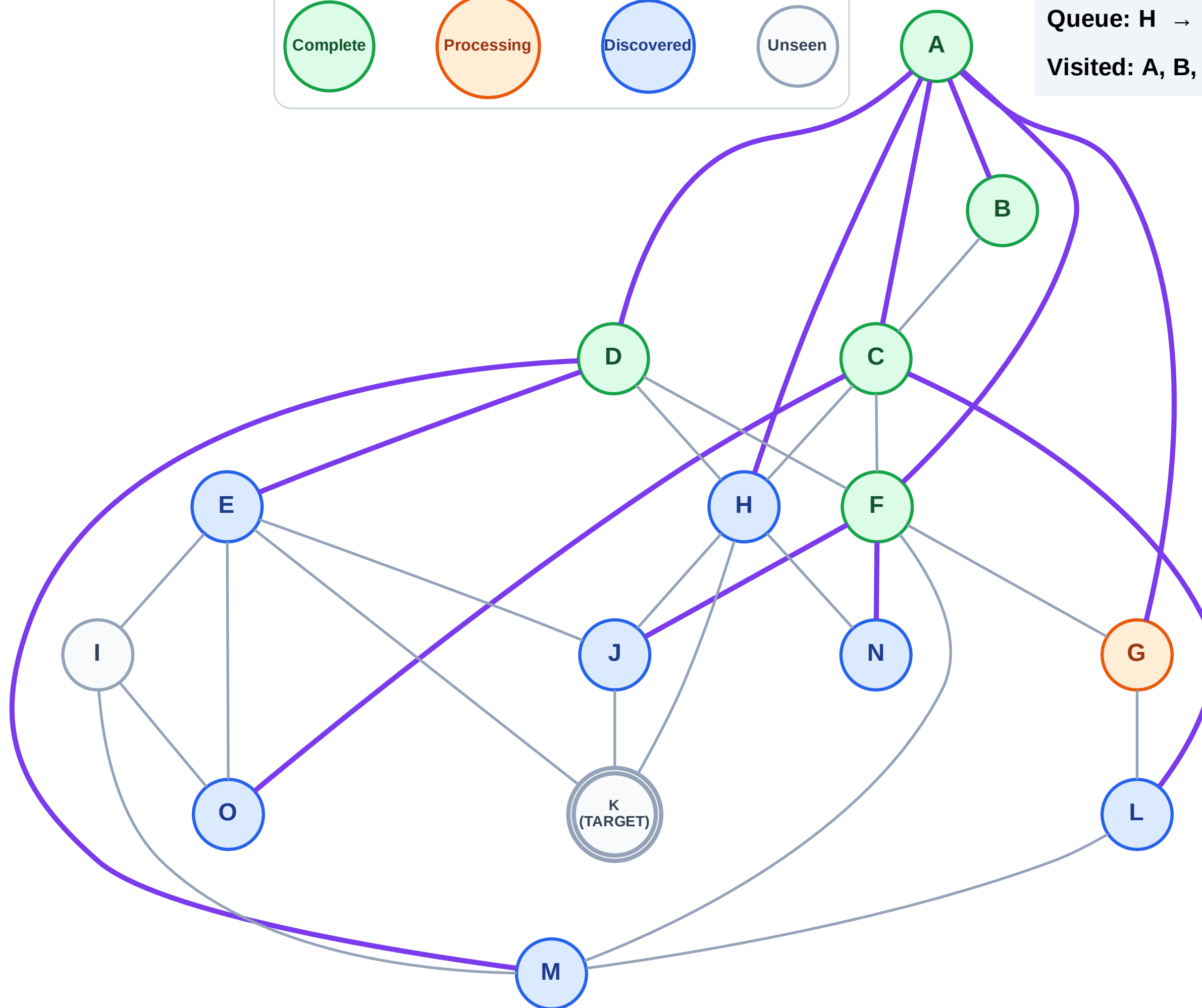
Step 12: Process node G

Legend



Queue: H → L → O → E → M → J → N

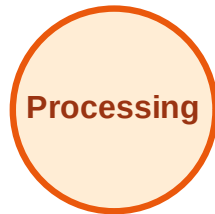
Visited: A, B, C, D, F



BFS Traversal

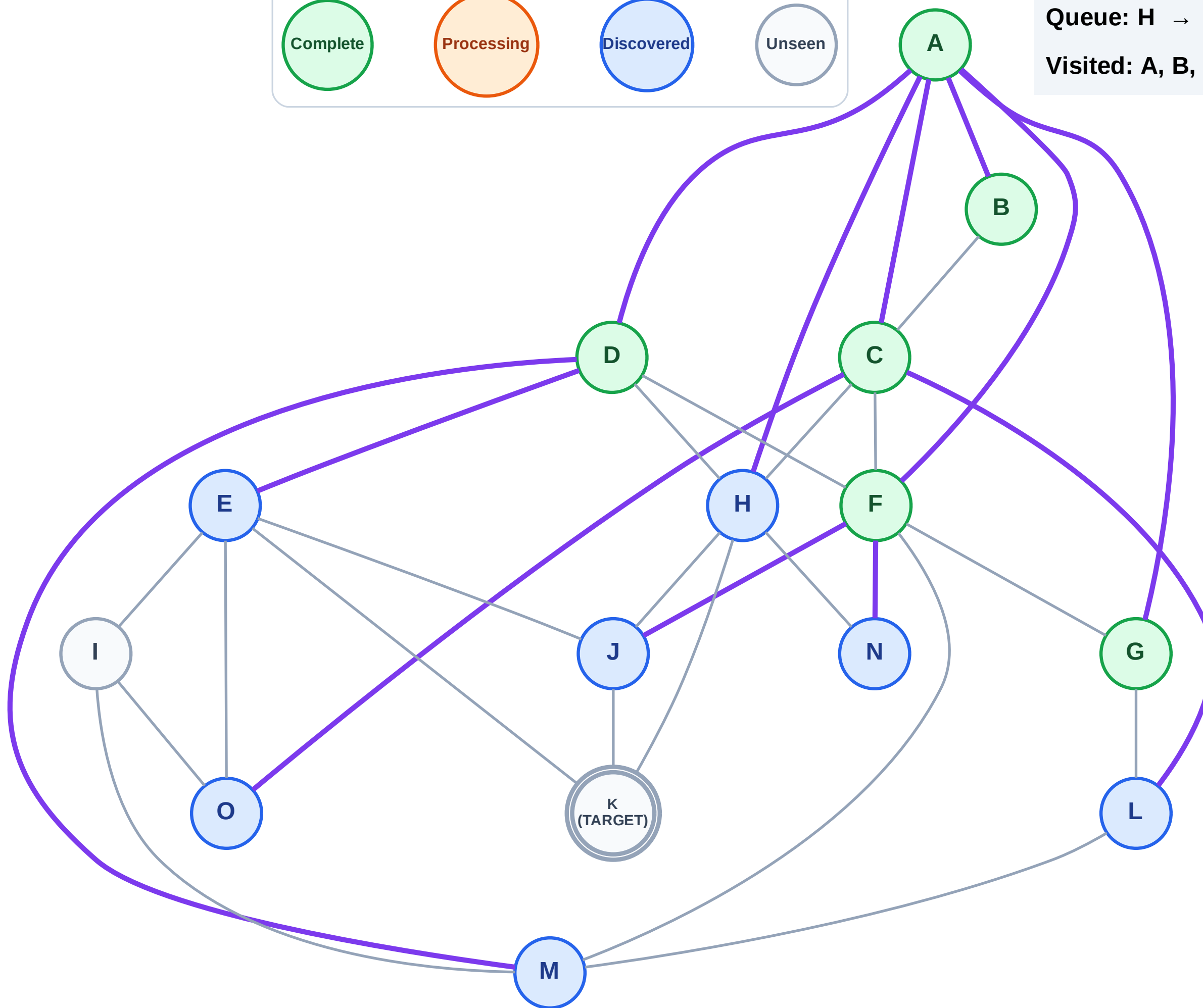
Step 13: No new neighbors from G

Legend



Queue: H → L → O → E → M → J → N

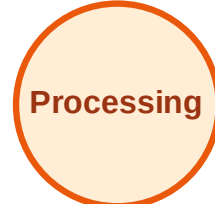
Visited: A, B, C, D, F, G



BFS Traversal

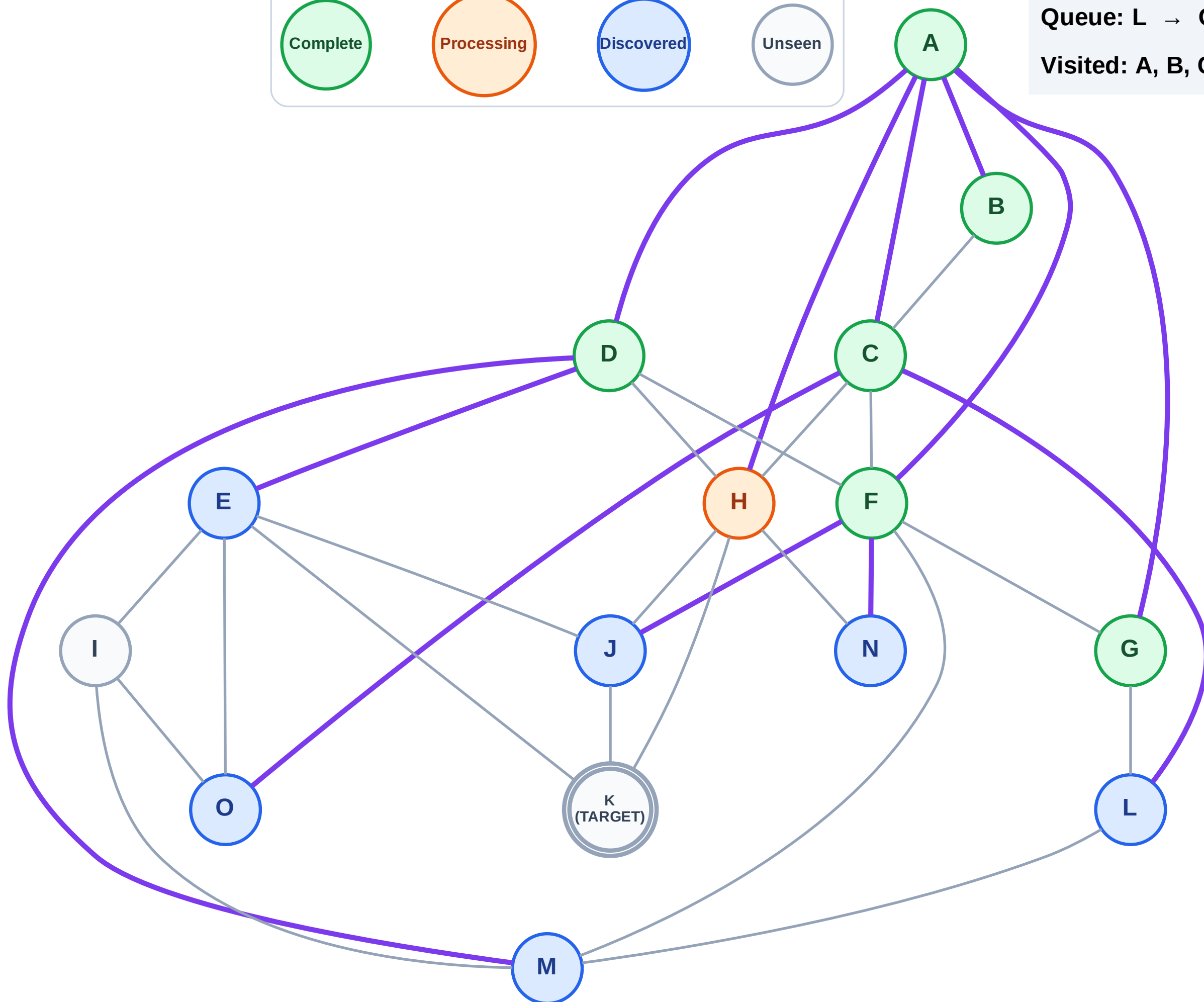
Step 14: Process node H

Legend



Queue: L → O → E → M → J → N

Visited: A, B, C, D, F, G



BFS Traversal

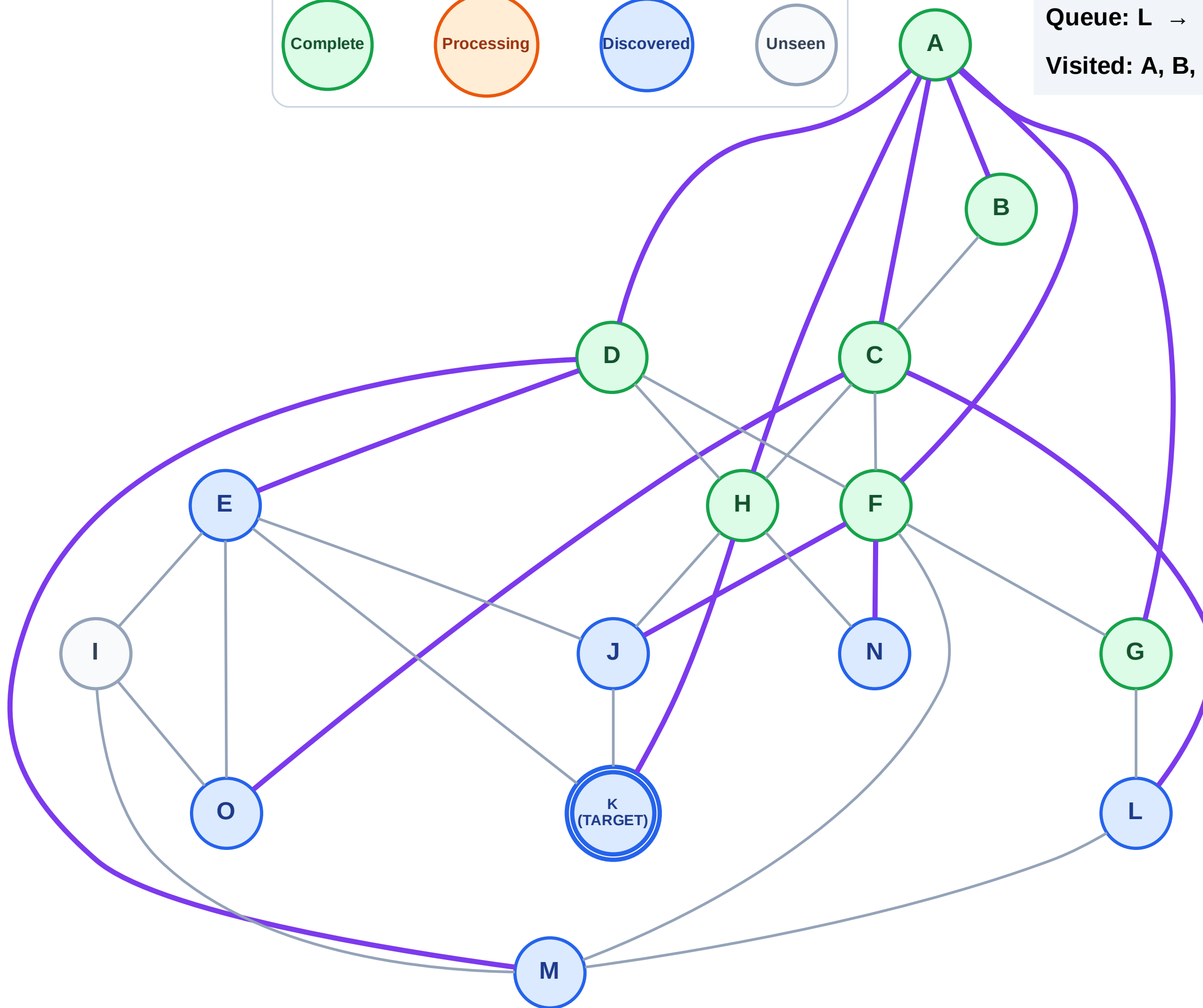
Step 15: Discover K from H

Legend



Queue: L → O → E → M → J → N → K

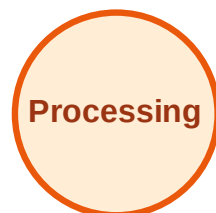
Visited: A, B, C, D, F, G, H



BFS Traversal

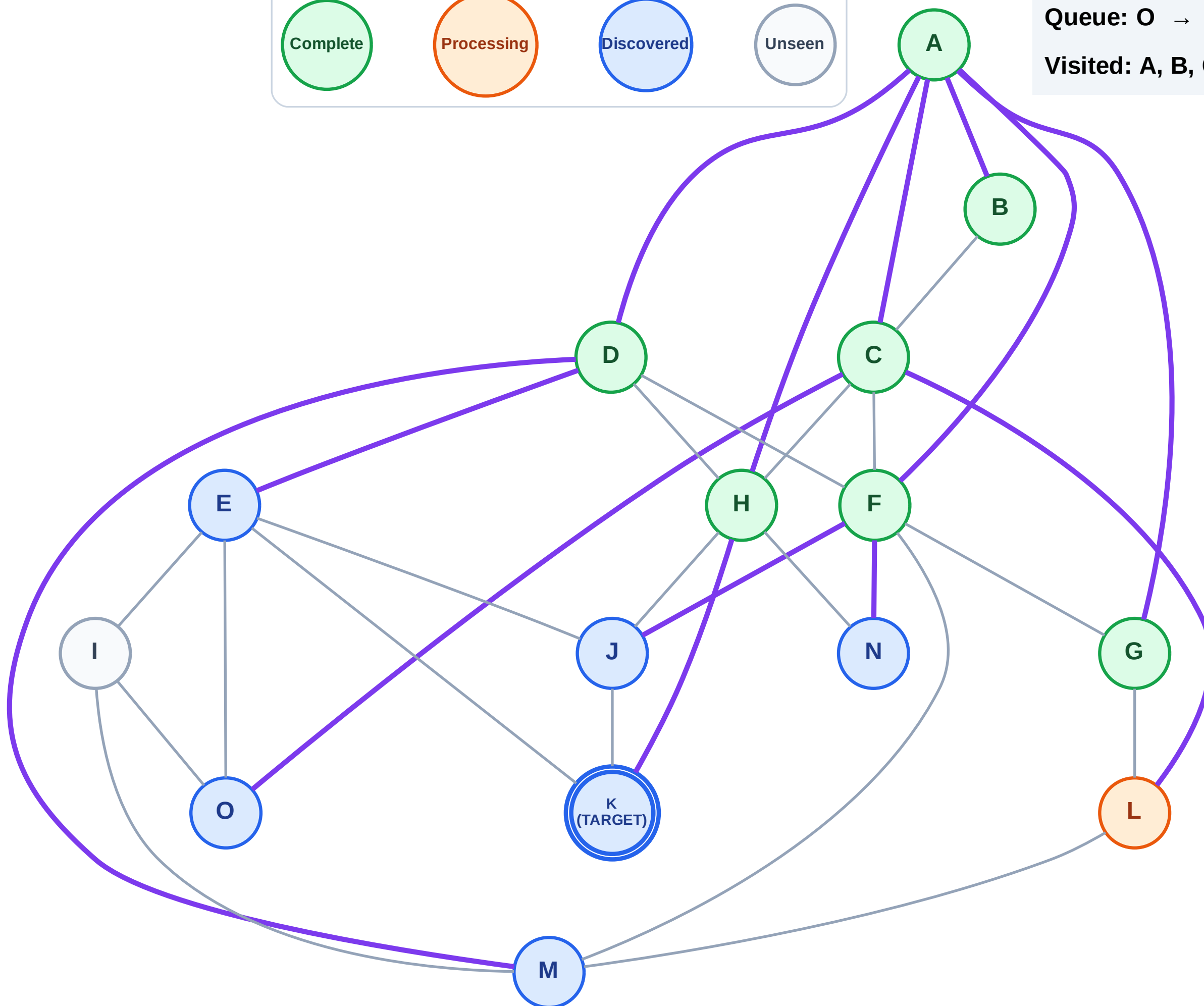
Step 16: Process node L

Legend



Queue: O → E → M → J → N → K

Visited: A, B, C, D, F, G, H



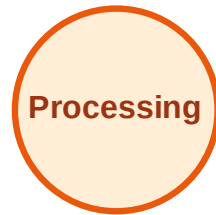
BFS Traversal

Step 17: No new neighbors from L

Legend



Complete



Processing



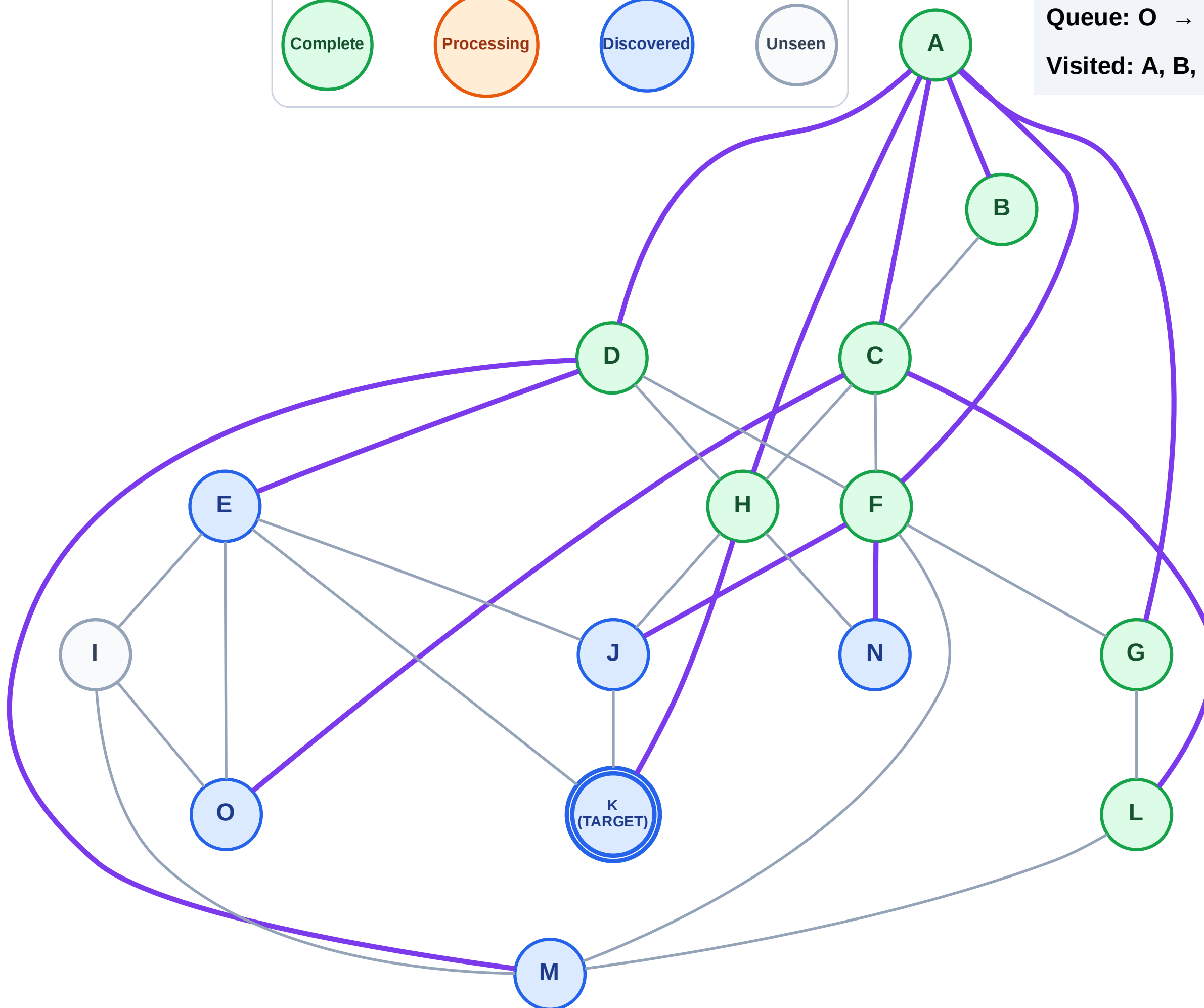
Discovered



Unseen

Queue: O → E → M → J → N → K

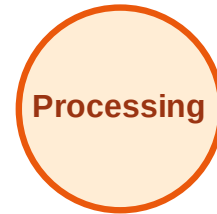
Visited: A, B, C, D, F, G, H, L



BFS Traversal

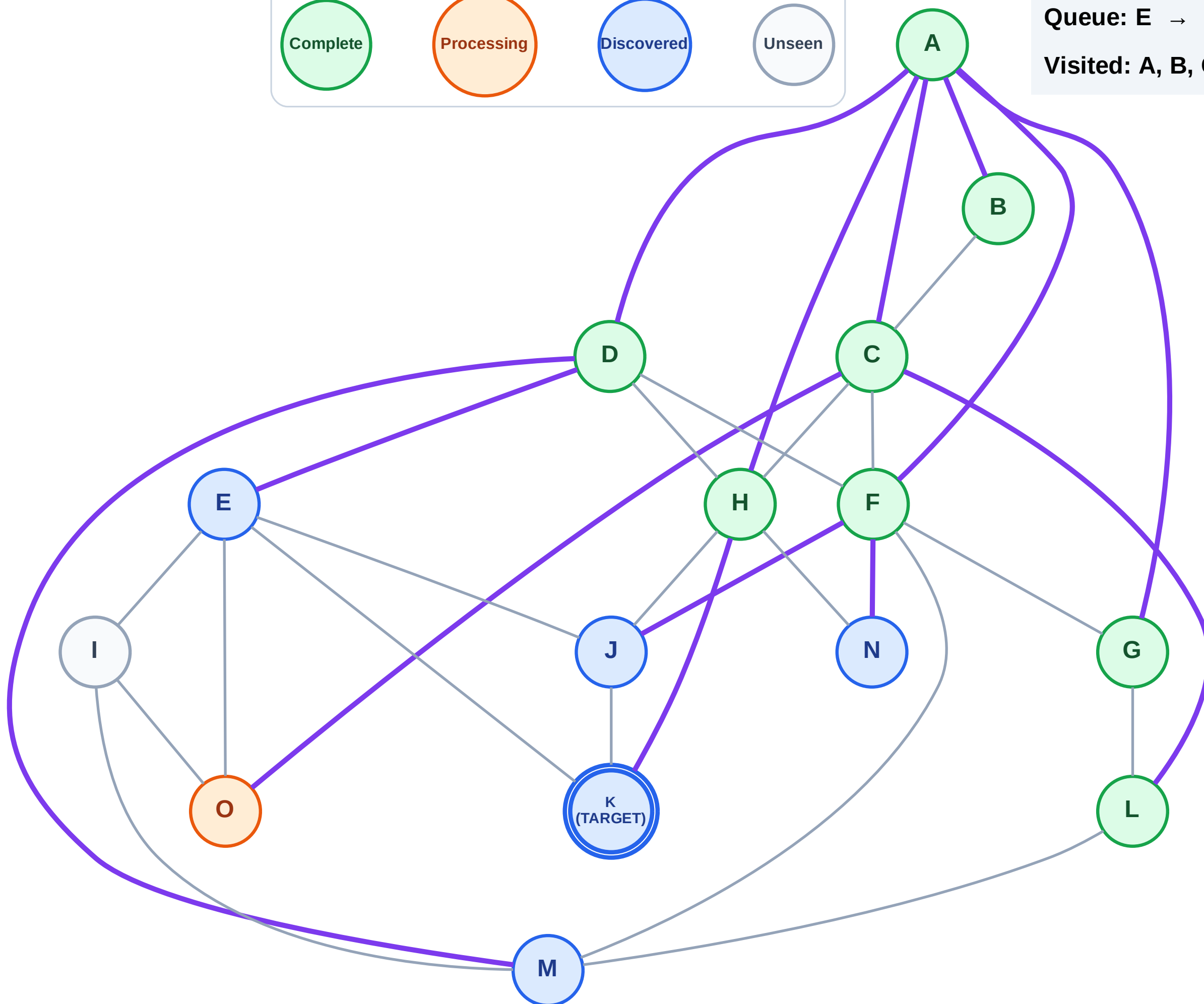
Step 18: Process node O

Legend



Queue: E → M → J → N → K

Visited: A, B, C, D, F, G, H, L



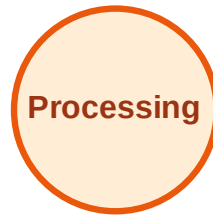
BFS Traversal

Step 19: Discover I from O

Legend



Complete



Processing



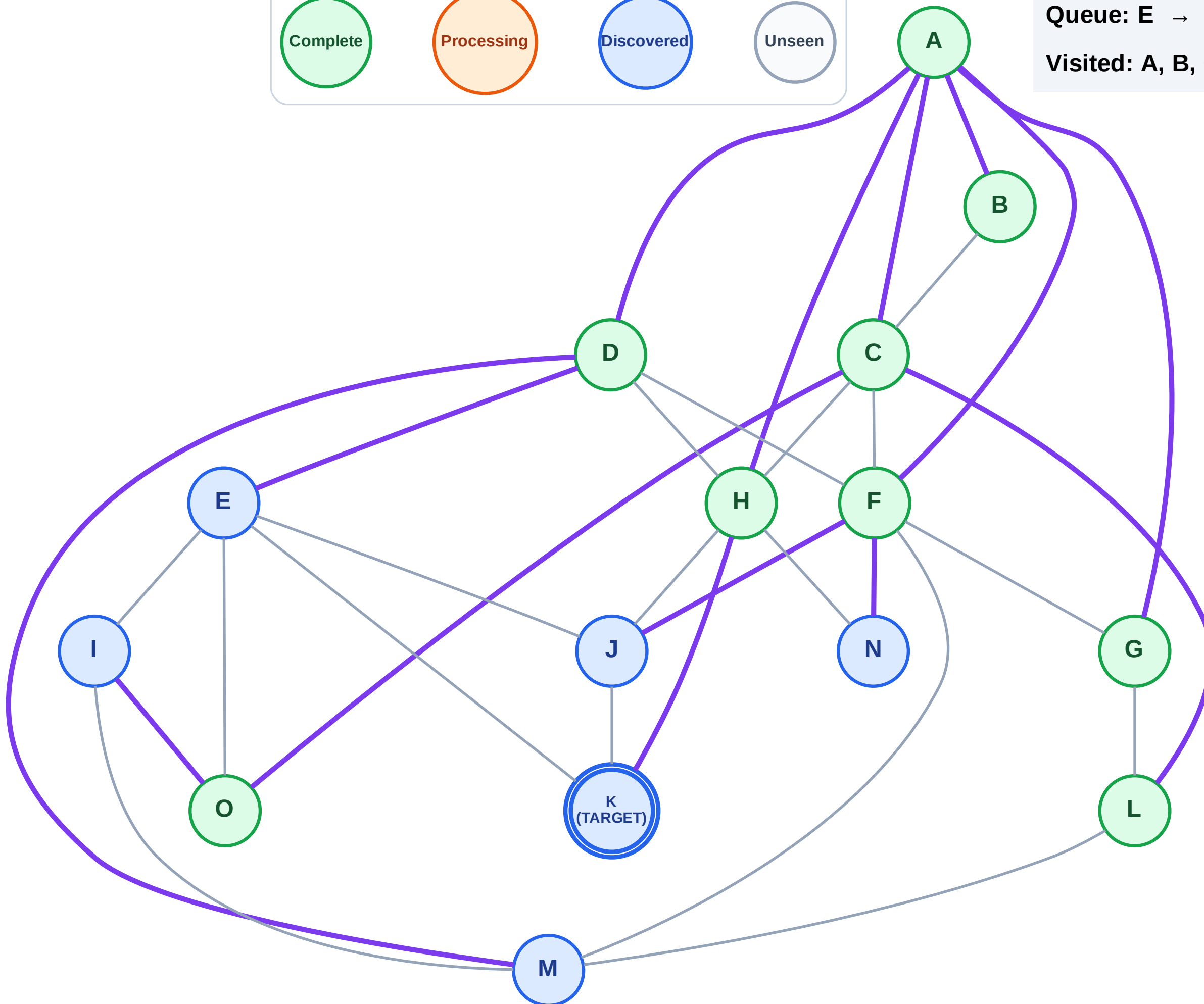
Discovered



Unseen

Queue: E → M → J → N → K → I

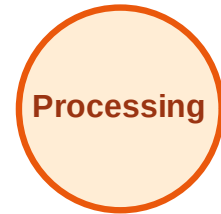
Visited: A, B, C, D, F, G, H, L, O



BFS Traversal

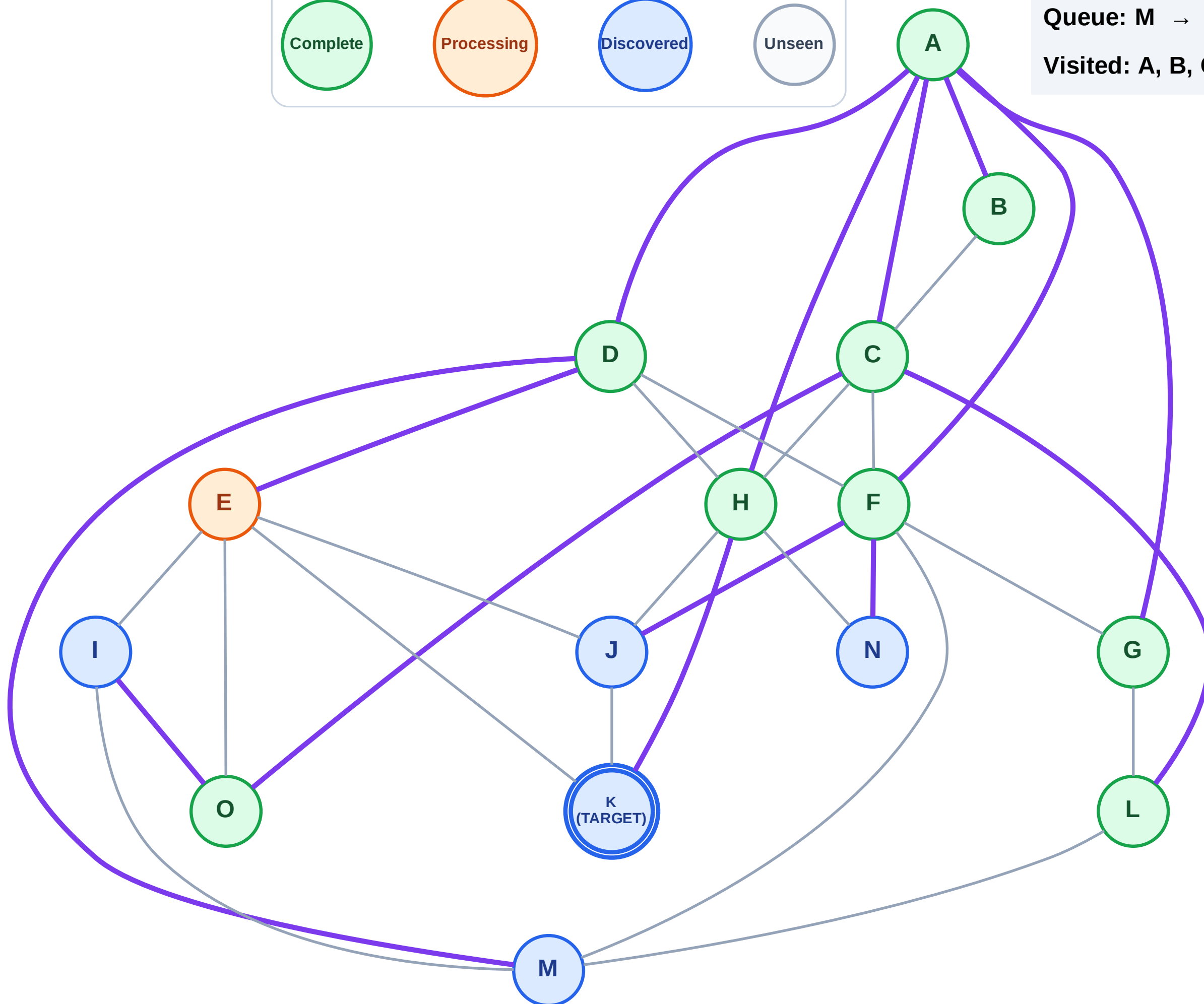
Step 20: Process node E

Legend



Queue: M → J → N → K → I

Visited: A, B, C, D, F, G, H, L, O



BFS Traversal

Step 21: No new neighbors from E

Legend

Complete

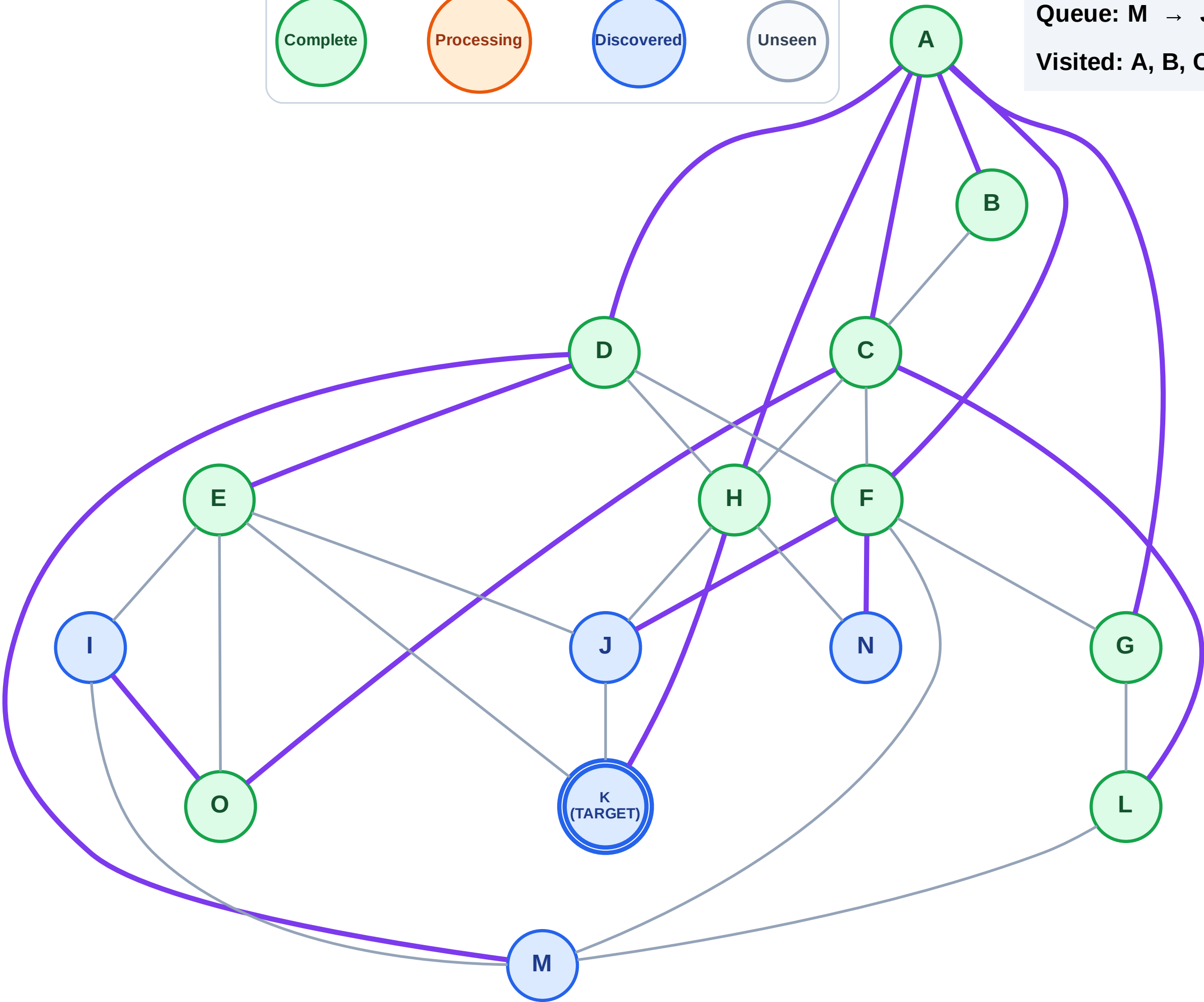
Processing

Discovered

Unseen

Queue: M → J → N → K → I

Visited: A, B, C, D, E, F, G, H, L, O



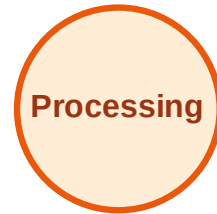
BFS Traversal

Step 22: Process node M

Legend



Complete



Processing



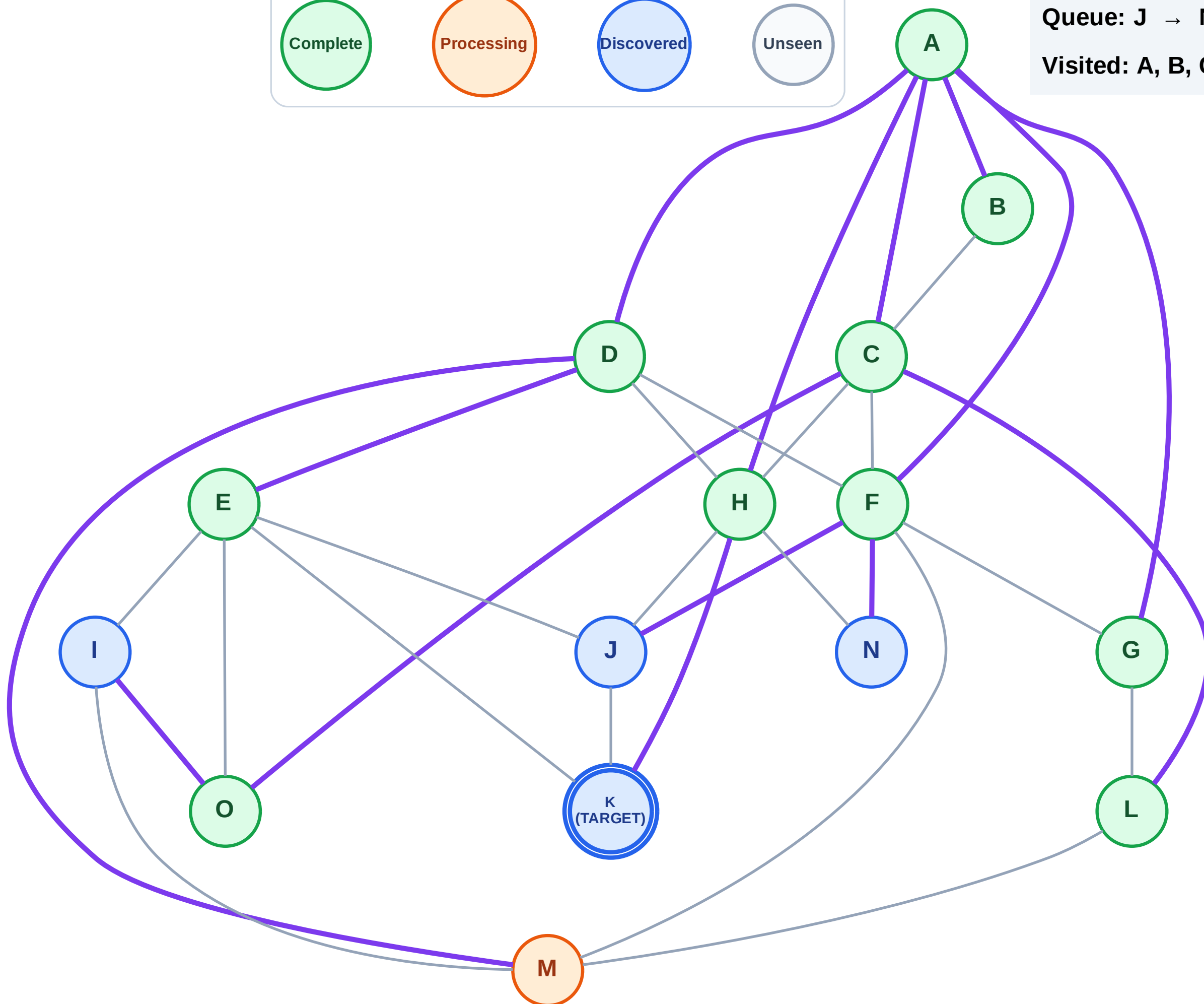
Discovered



Unseen

Queue: J → N → K → I

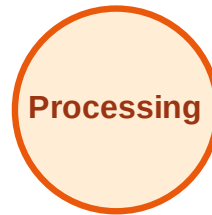
Visited: A, B, C, D, E, F, G, H, L, O



BFS Traversal

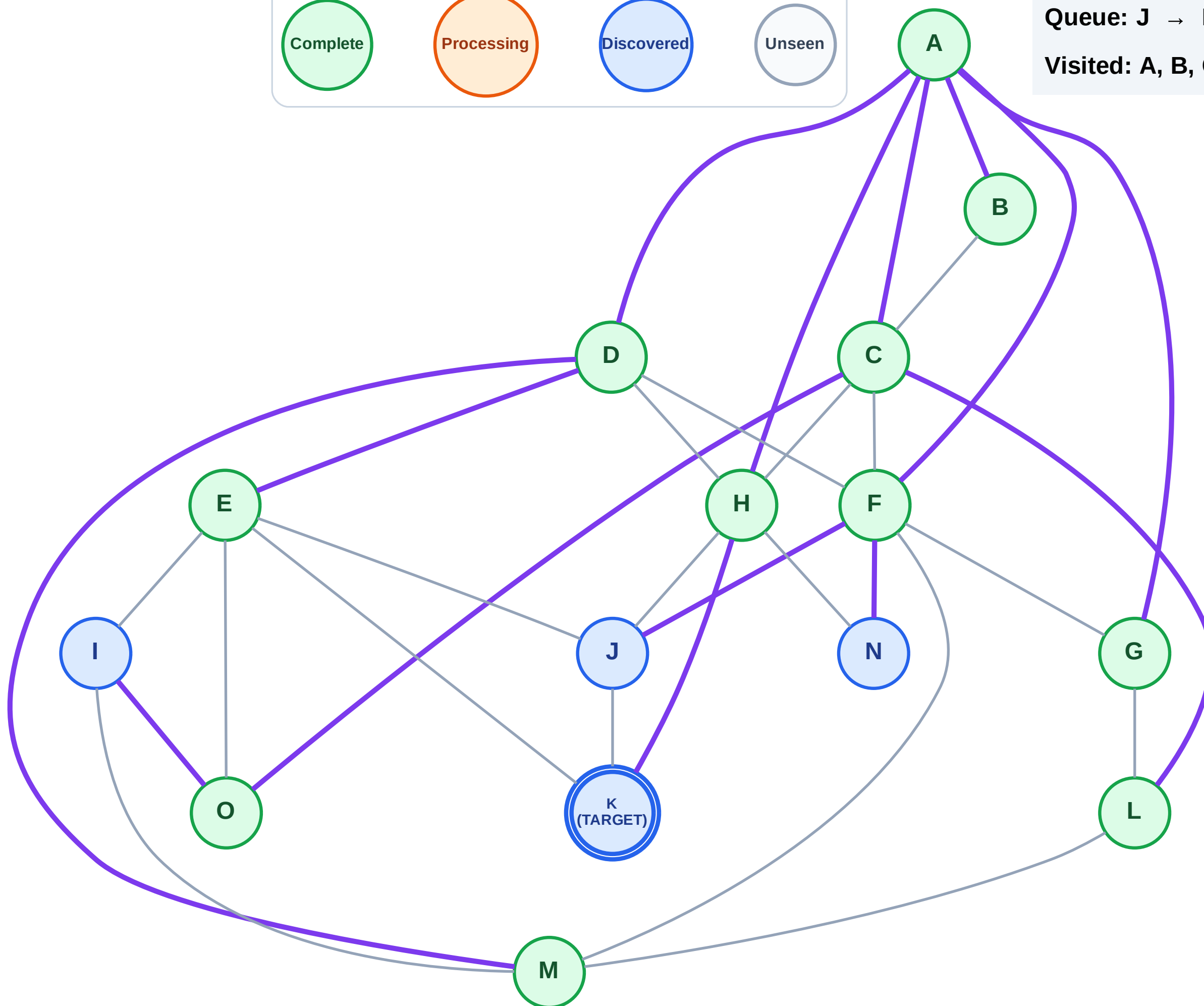
Step 23: No new neighbors from M

Legend



Queue: J → N → K → I

Visited: A, B, C, D, E, F, G, H, L, M, O



BFS Traversal

Step 24: Process node J

Legend

Complete

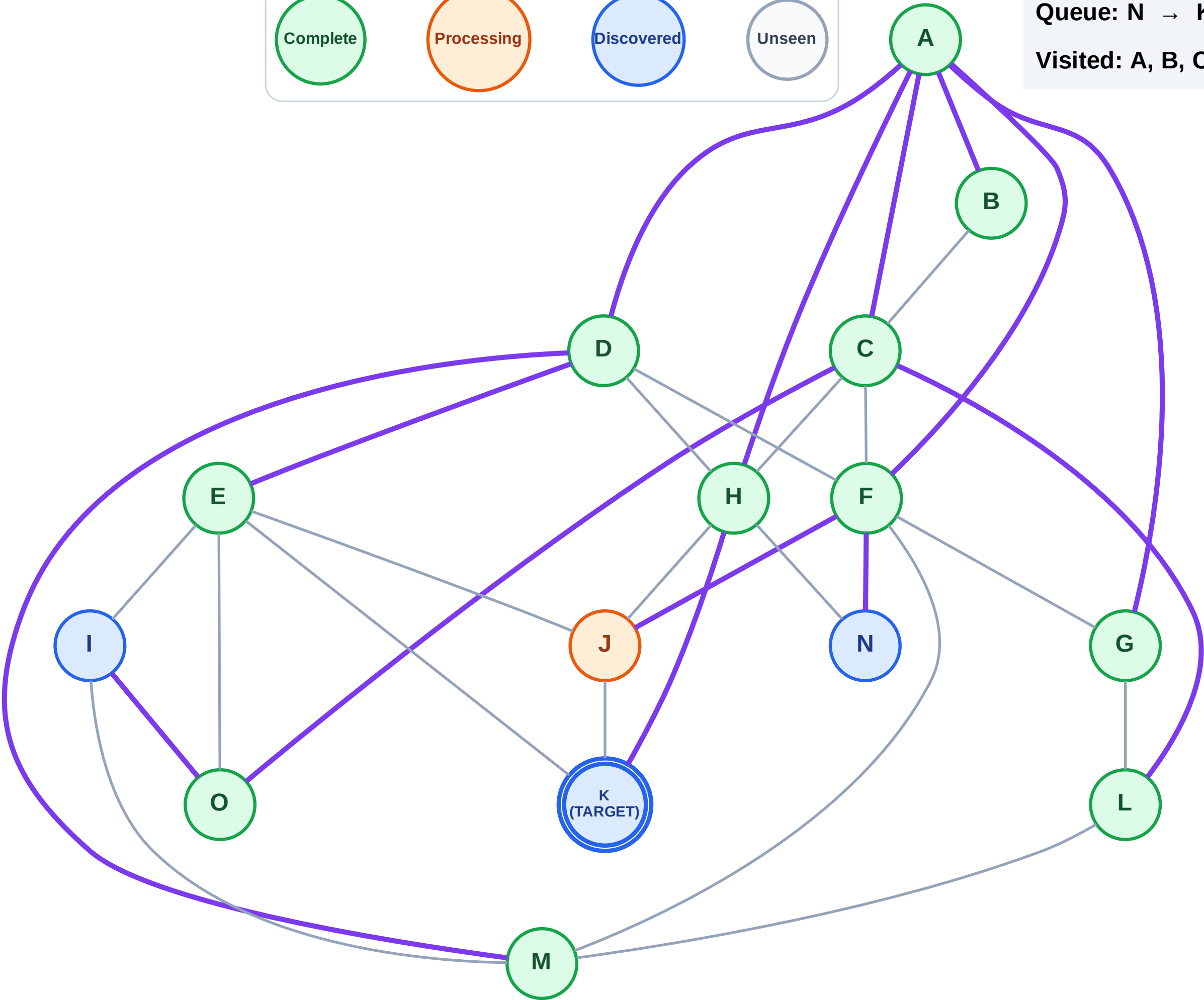
Processing

Discovered

Unseen

Queue: N → K → I

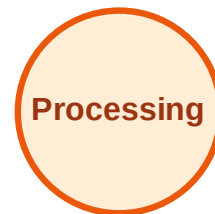
Visited: A, B, C, D, E, F, G, H, L, M, O



BFS Traversal

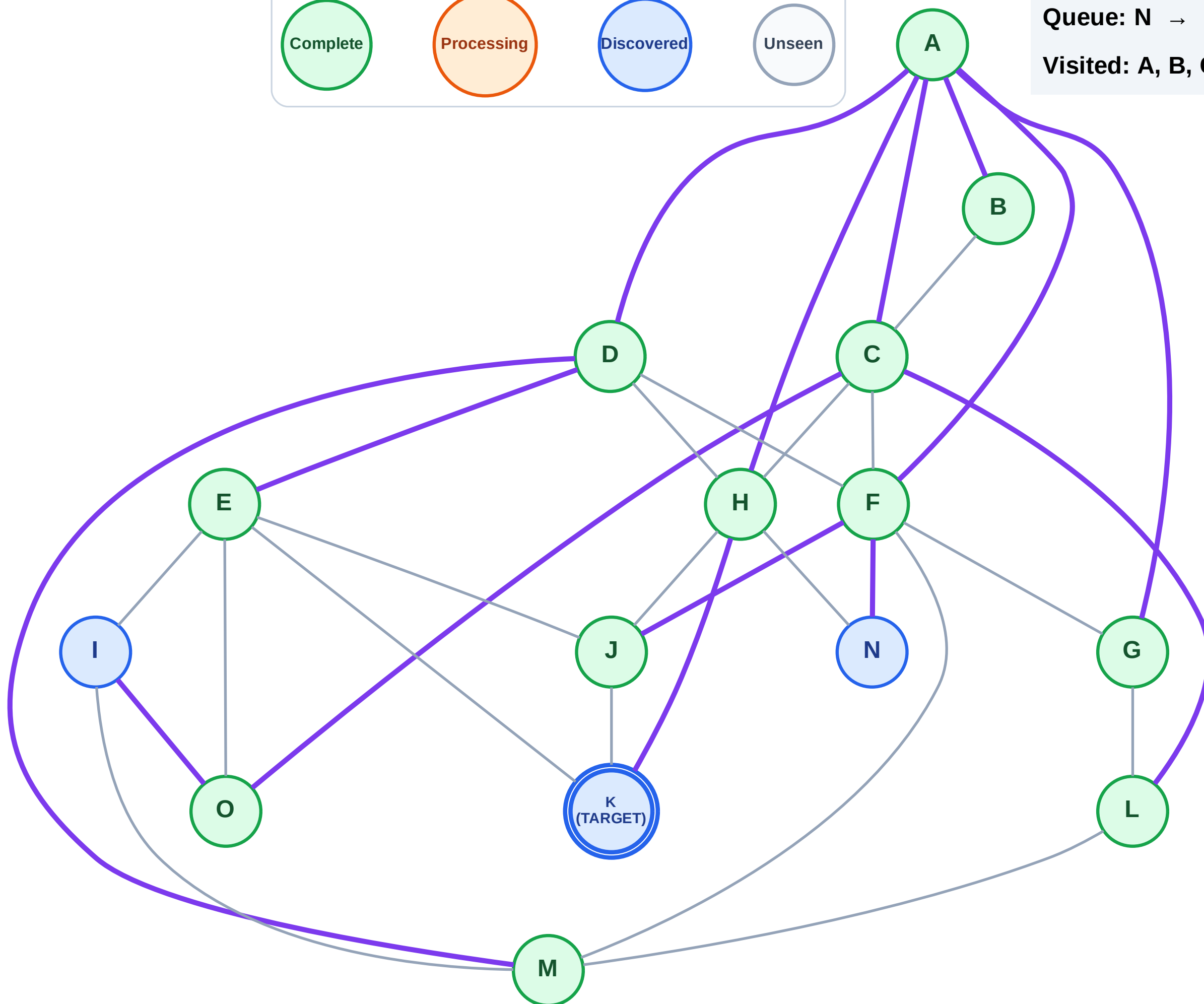
Step 25: No new neighbors from J

Legend



Queue: N → K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, O



BFS Traversal

Step 26: Process node N

Legend

Complete

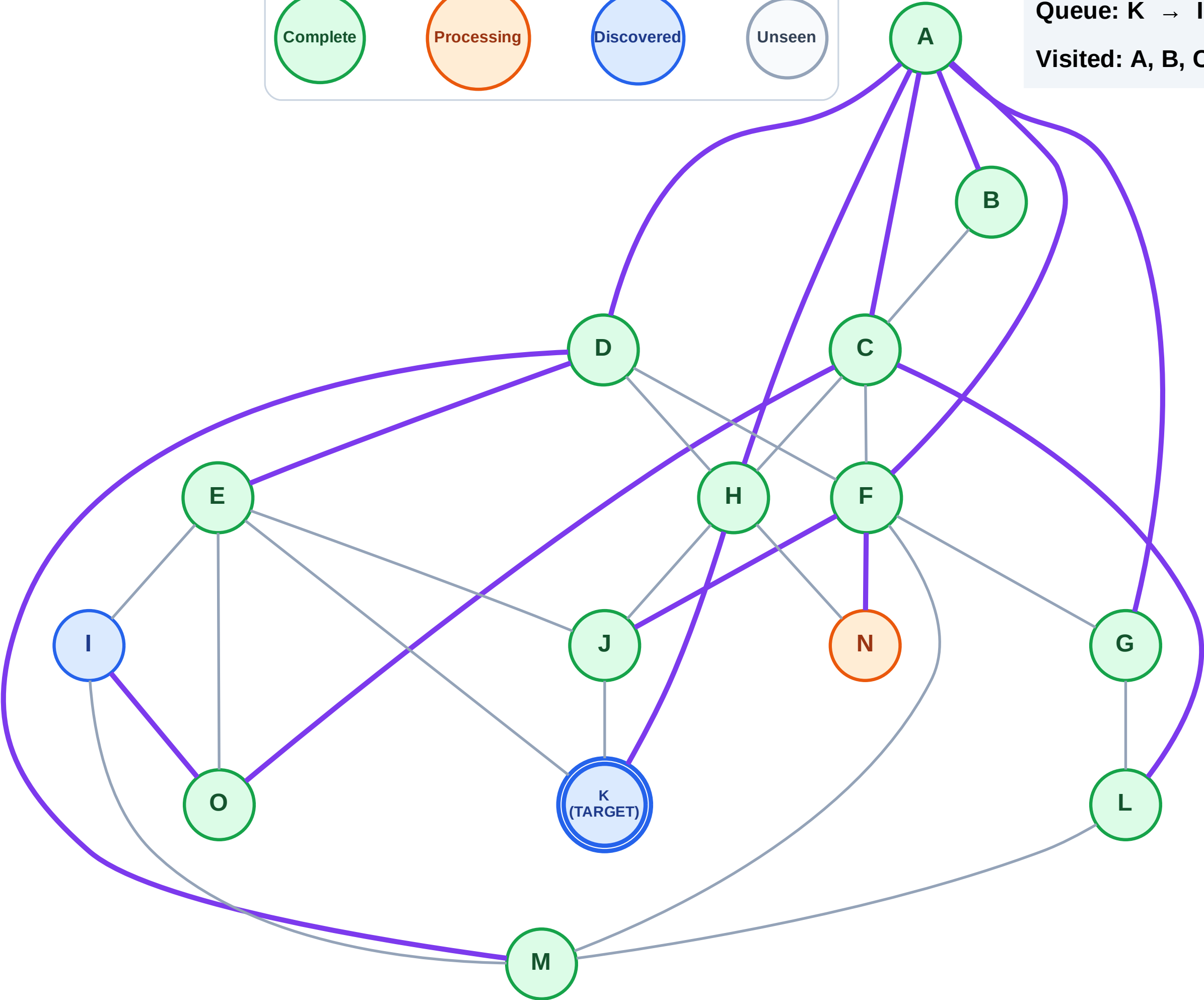
Processing

Discovered

Unseen

Queue: K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, O



BFS Traversal

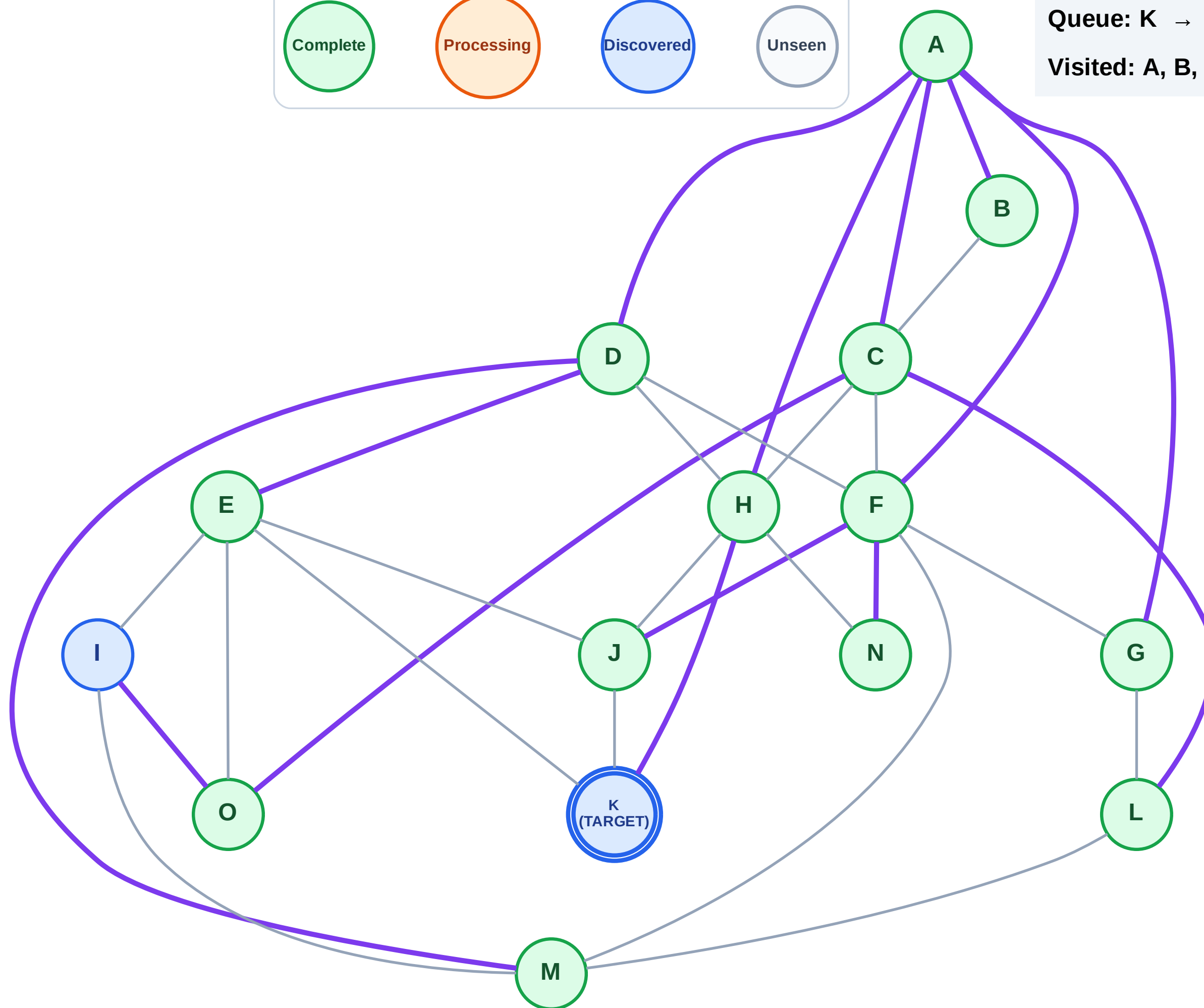
Step 27: No new neighbors from N

Legend



Queue: K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

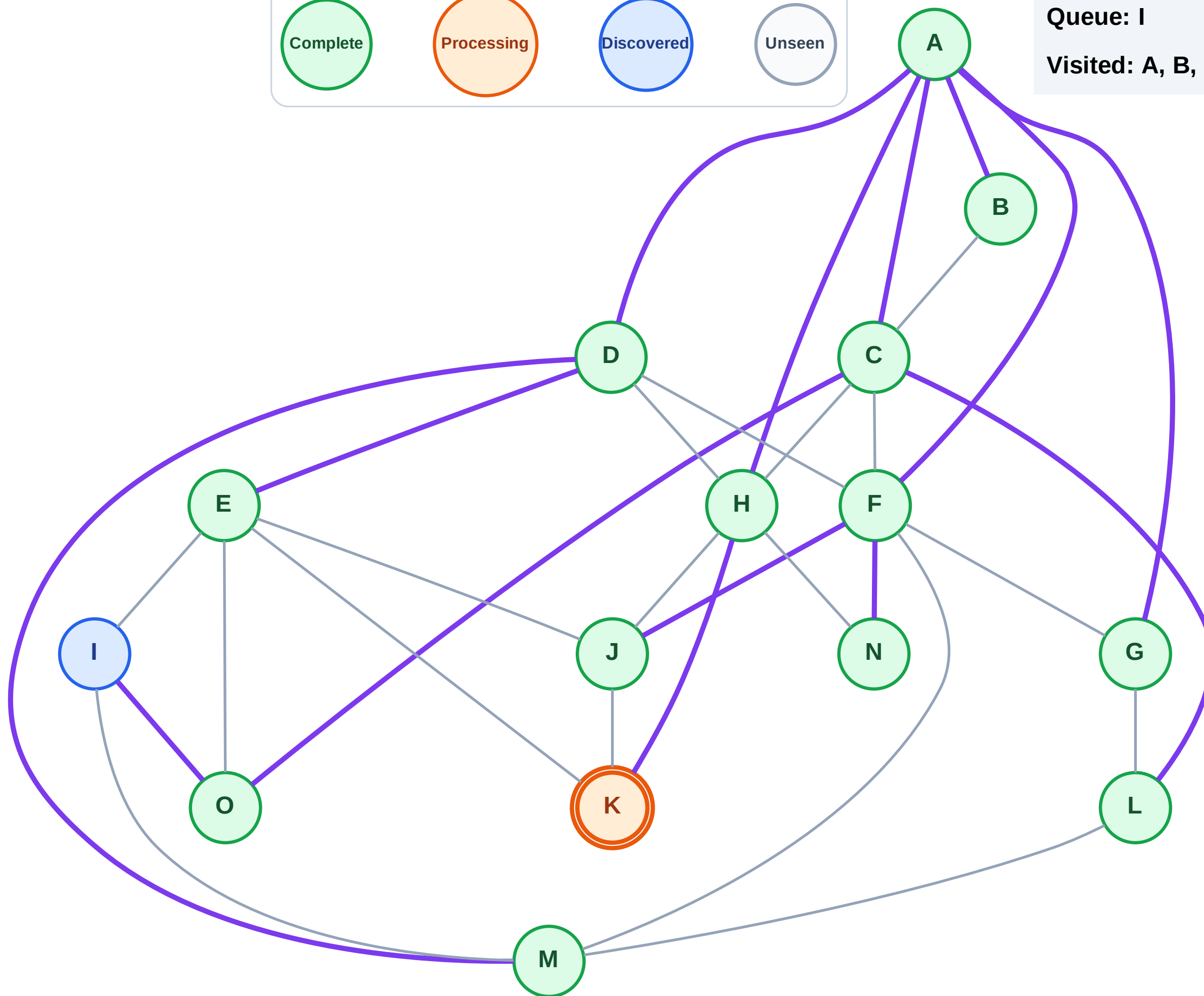
Step 28: Process node K

Legend



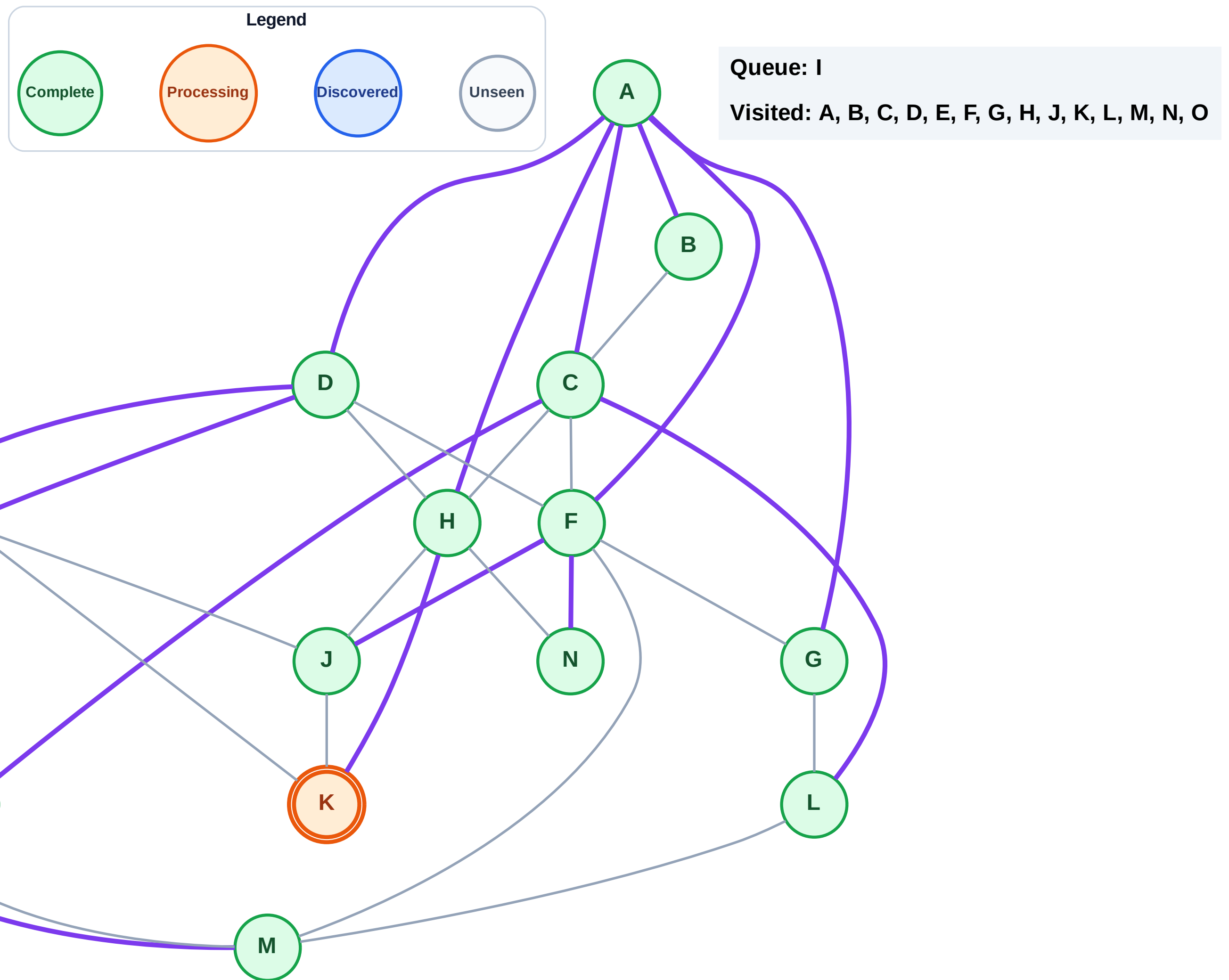
Queue: I

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

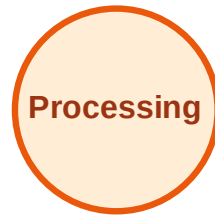
Step 29: TARGET NODE K REACHED! Path (2 hops): A → H → K



DFS Traversal

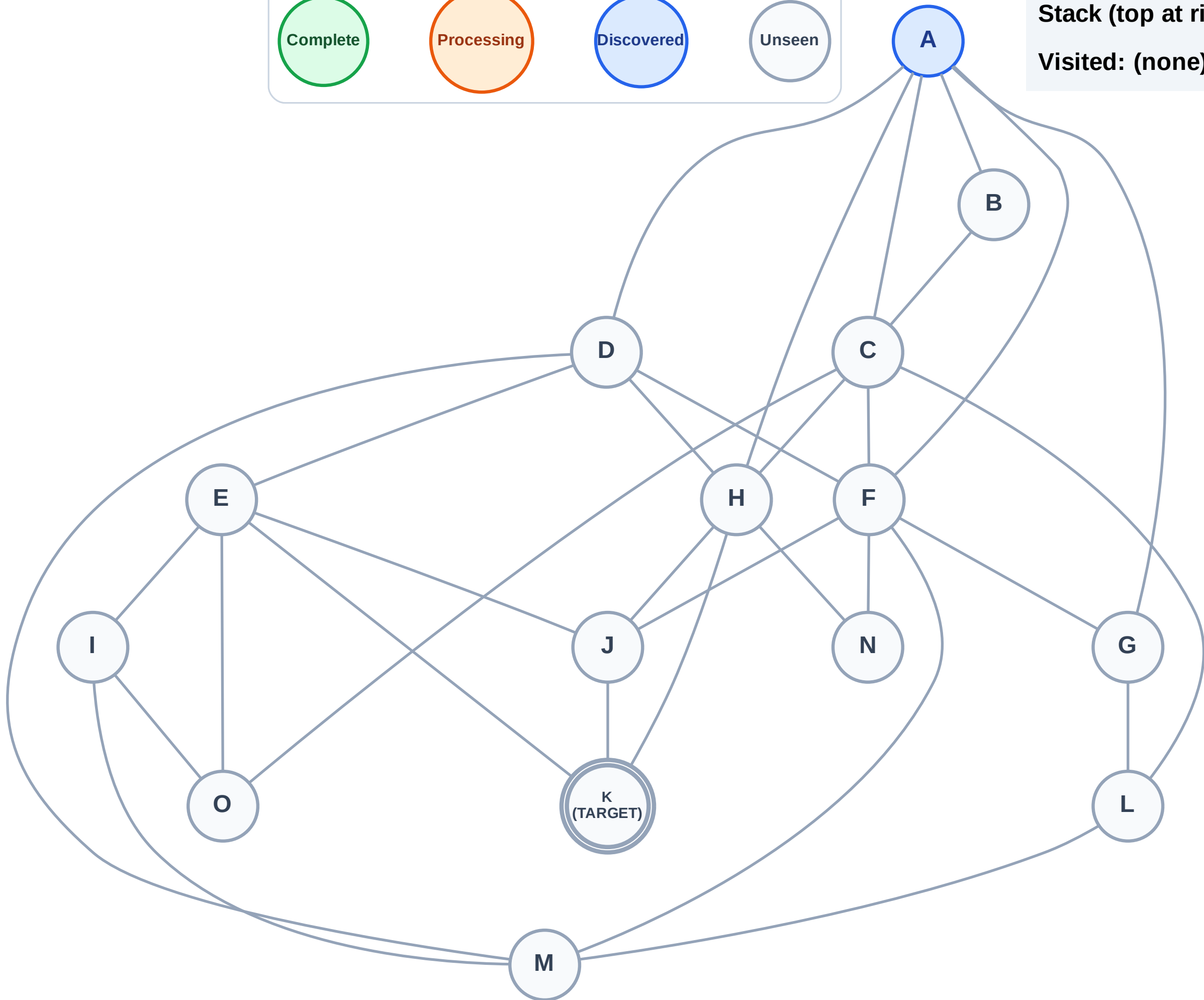
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

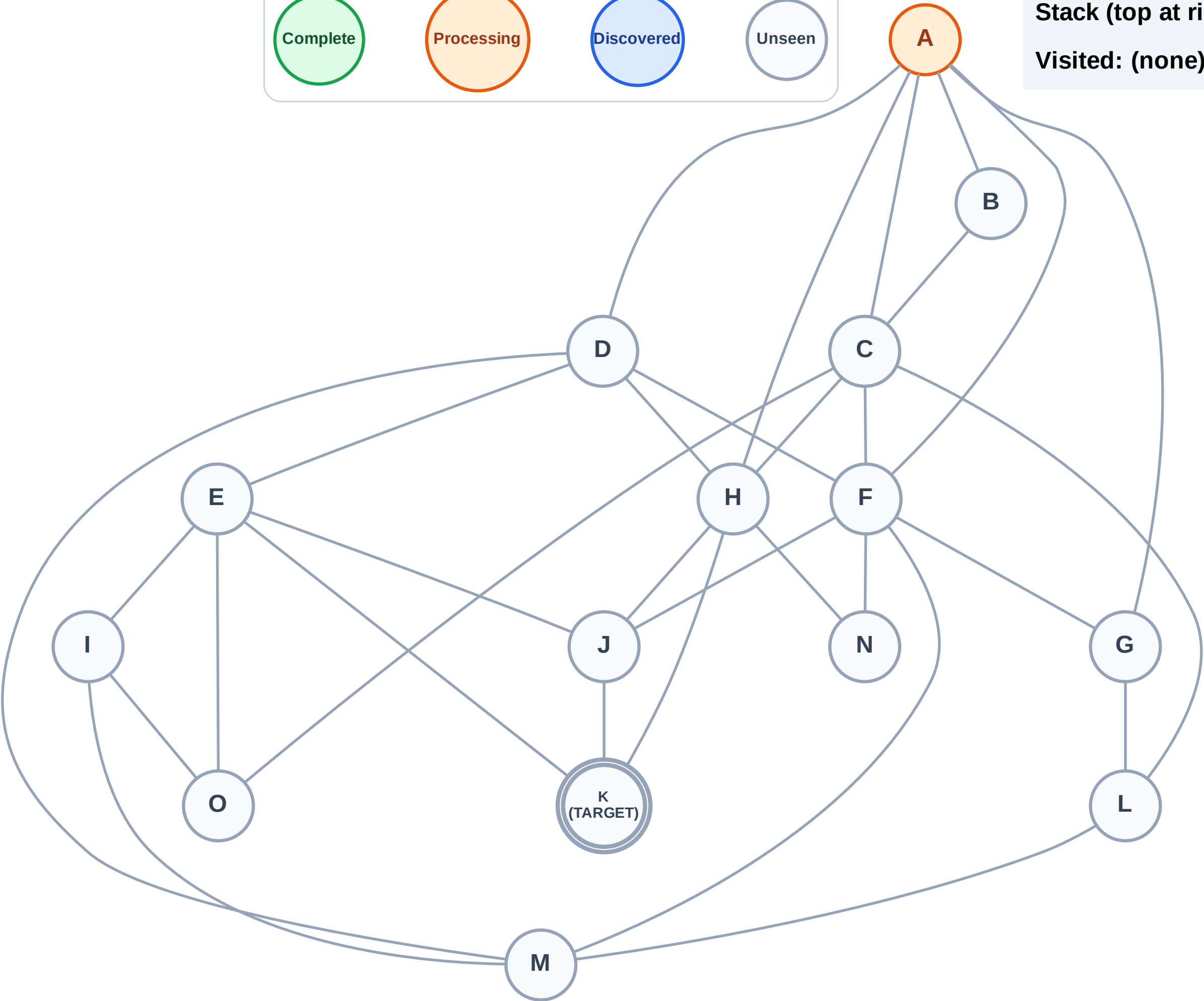
Complete

Processing

Discovered

Unseen

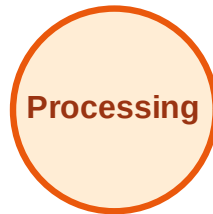
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

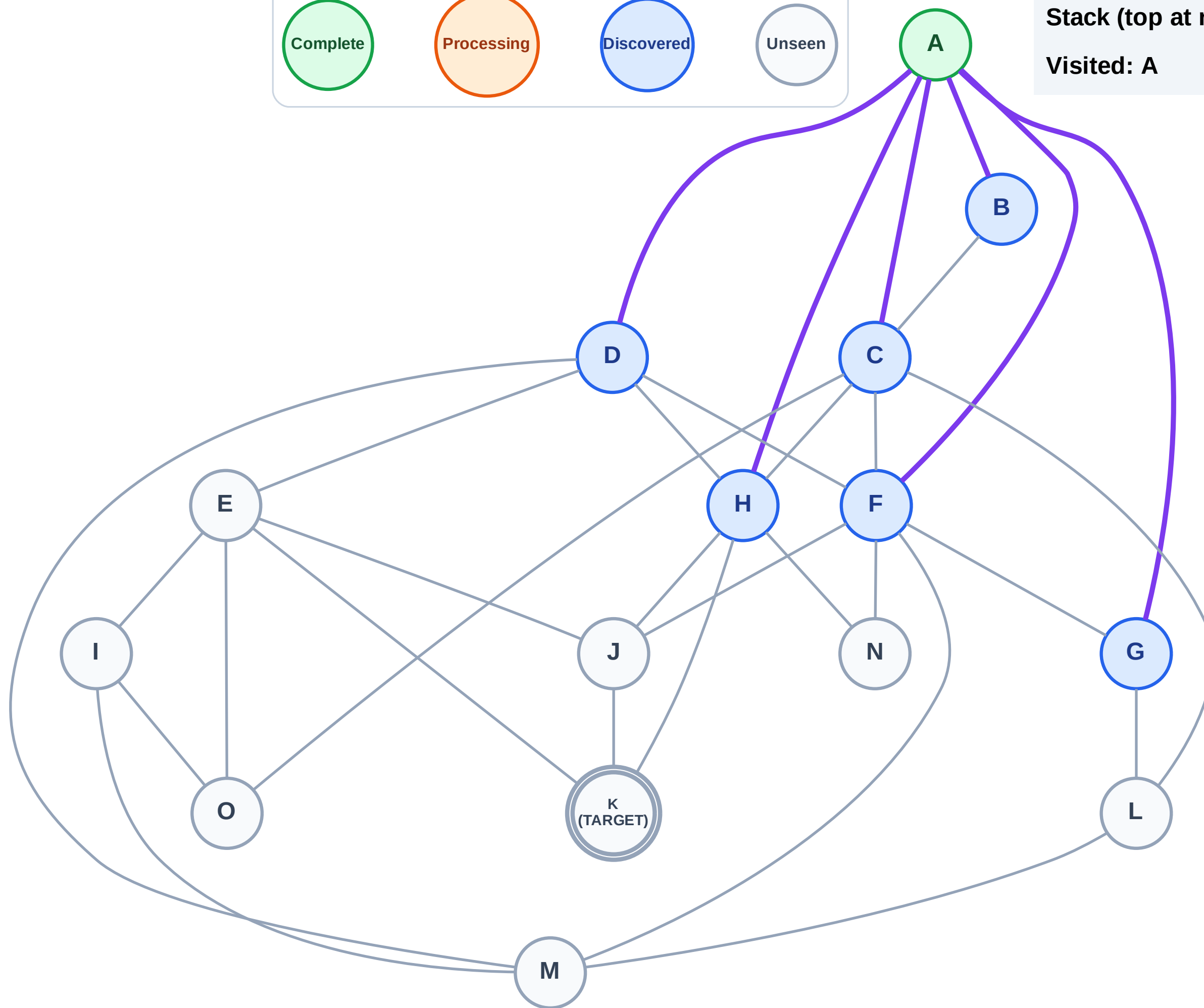
Step 3: Discover H, G, F, D, C, B from A

Legend



Stack (top at right): H | G | F | D | C | B

Visited: A



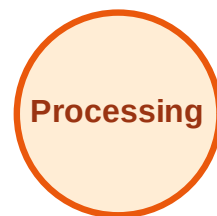
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



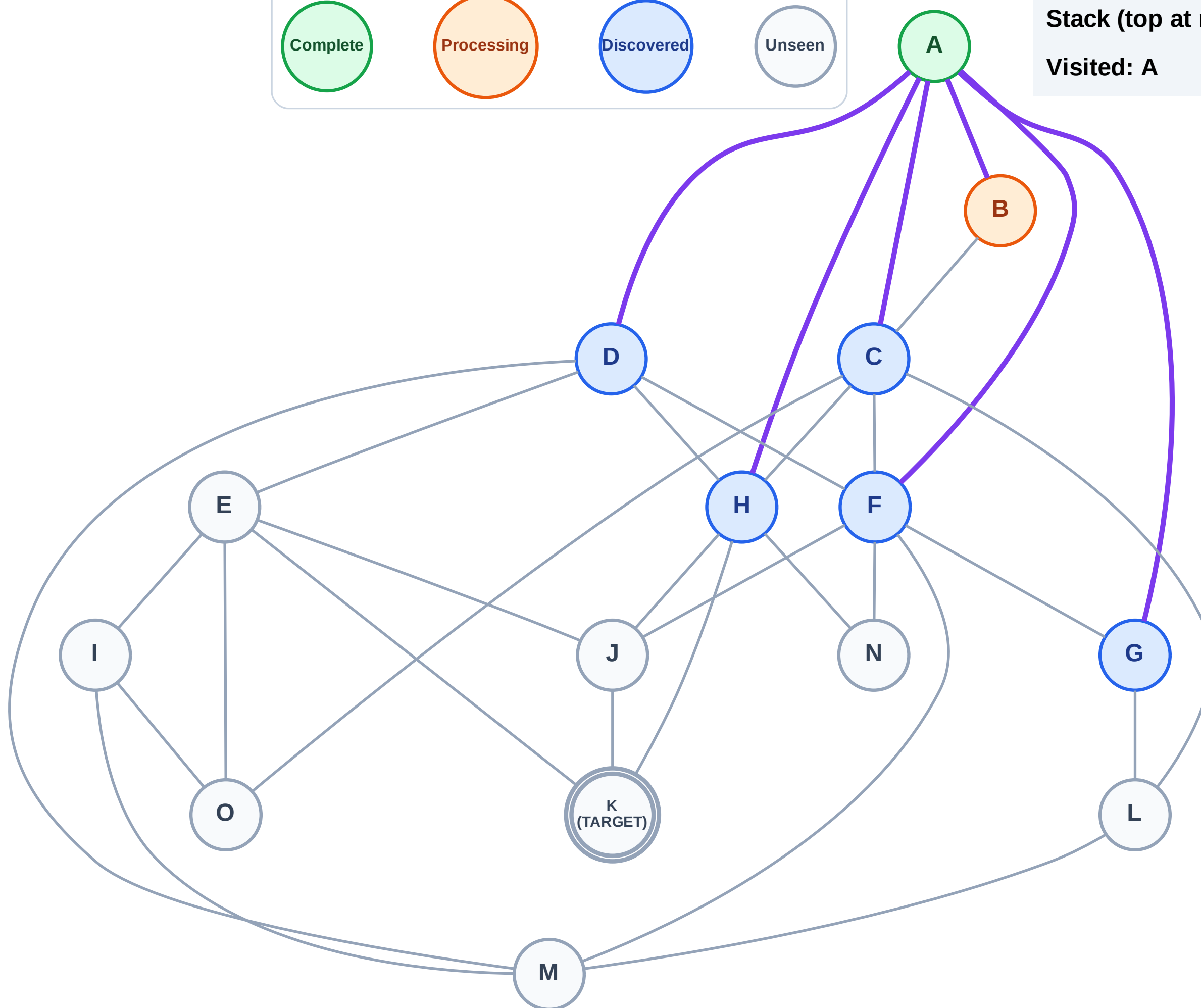
Discovered



Unseen

Stack (top at right): H | G | F | D | C

Visited: A



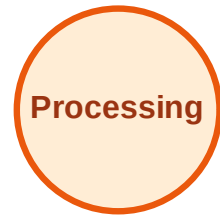
DFS Traversal

Step 5: No new neighbors from B

Legend



Complete



Processing



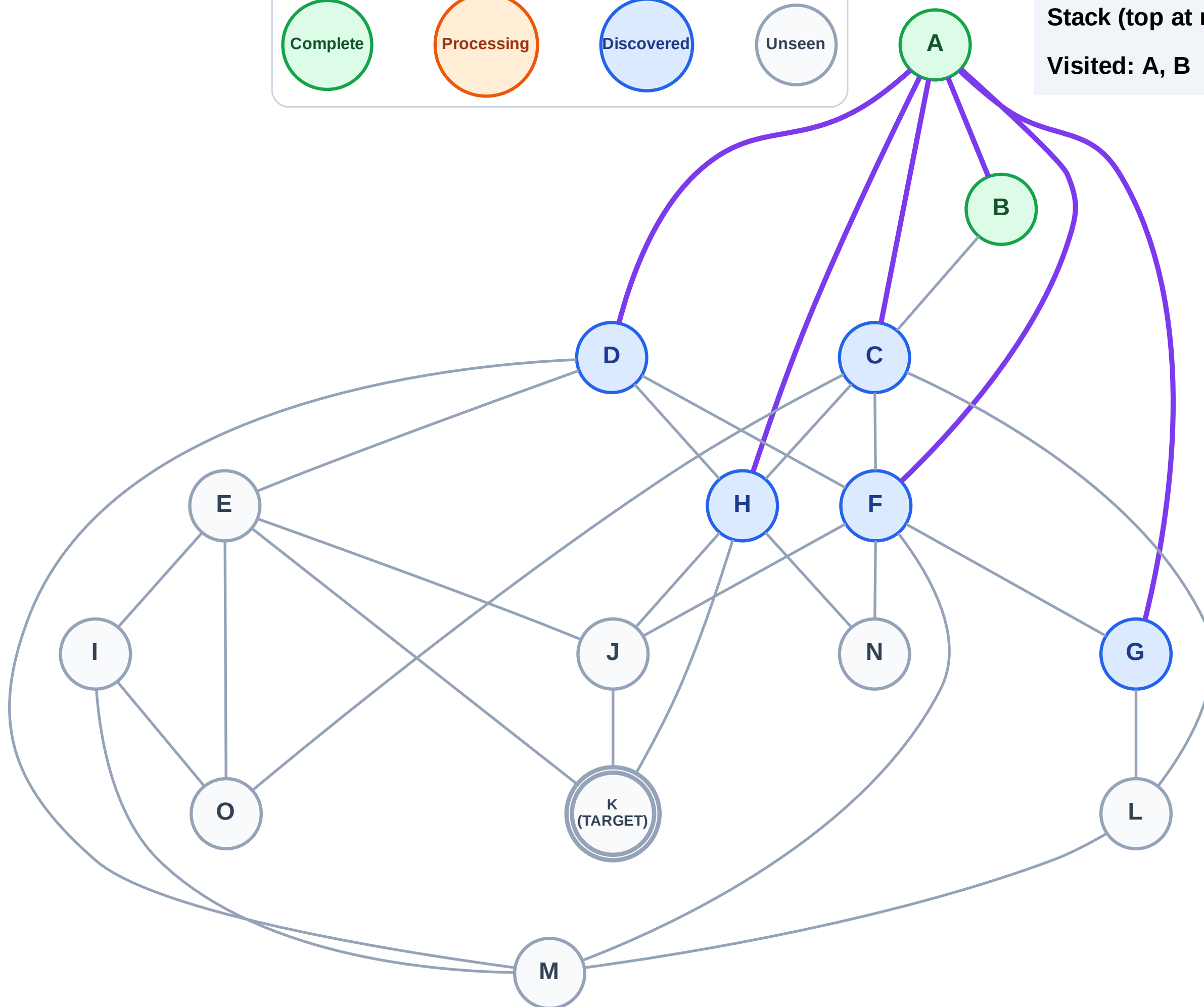
Discovered



Unseen

Stack (top at right): H | G | F | D | C

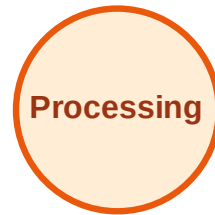
Visited: A, B



DFS Traversal

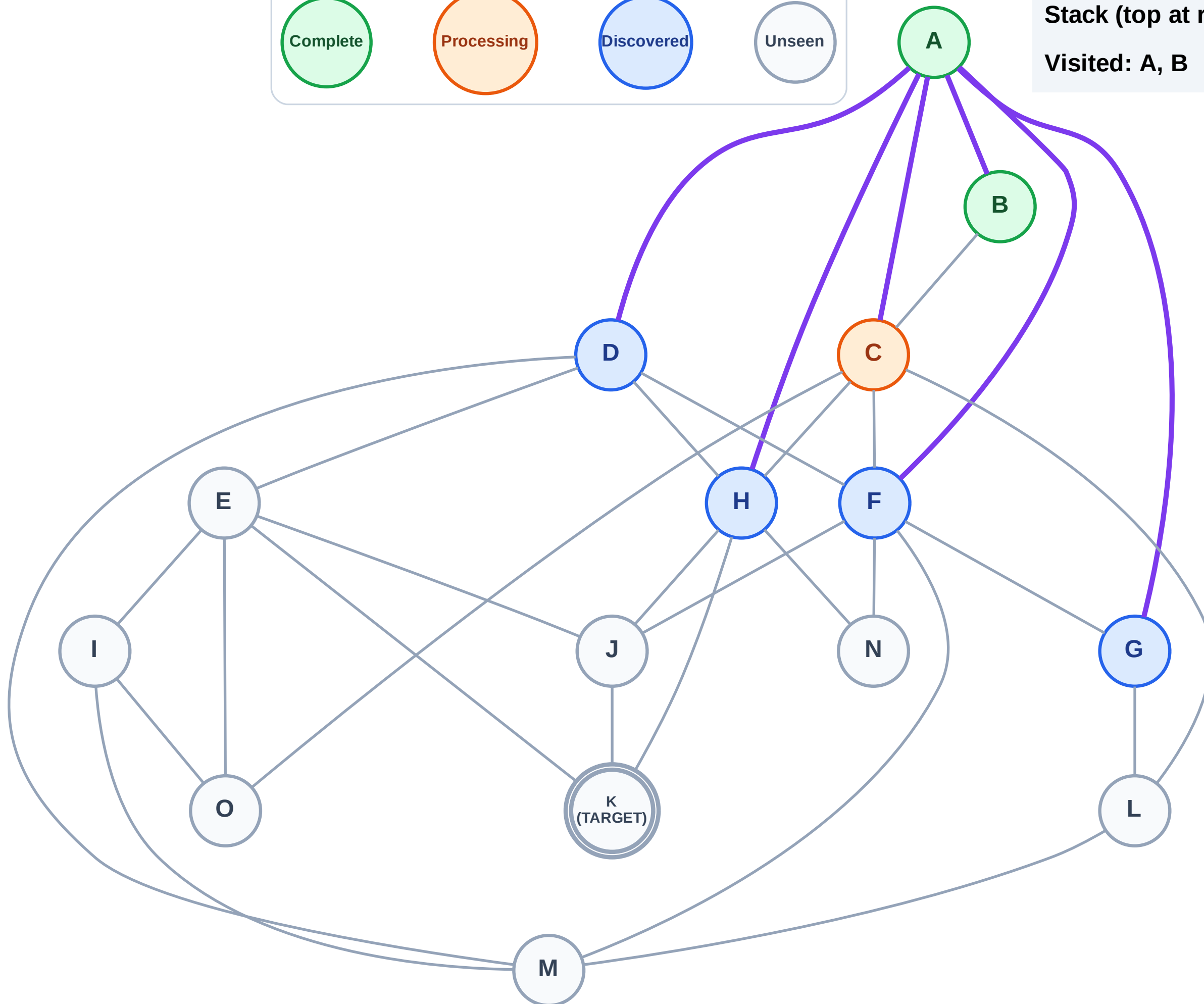
Step 6: Process node C

Legend



Stack (top at right): H | G | F | D

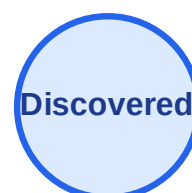
Visited: A, B



DFS Traversal

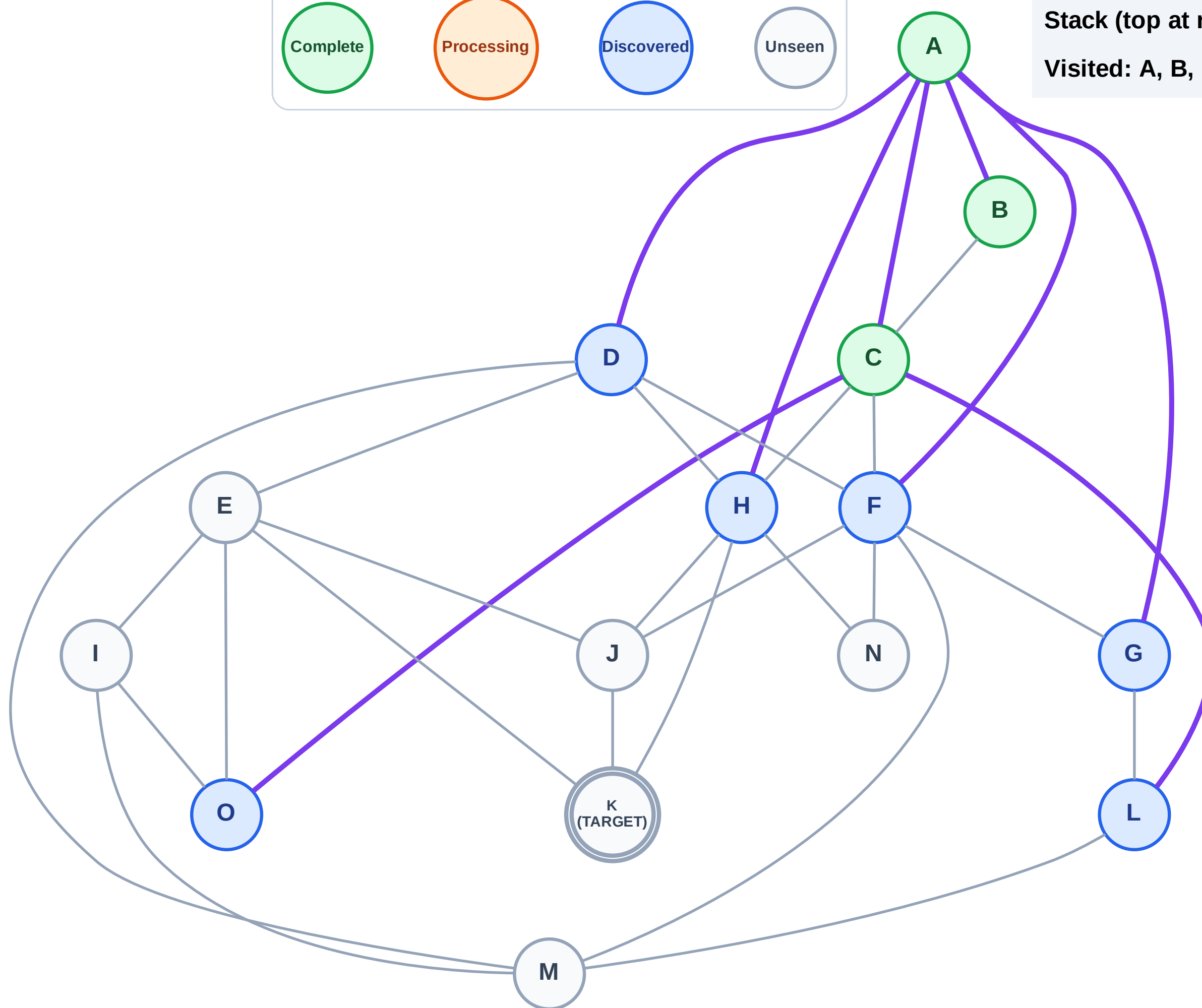
Step 7: Discover O, L from C

Legend



Stack (top at right): H | G | F | D | O | L

Visited: A, B, C



DFS Traversal

Step 8: Process node L

Legend

Complete

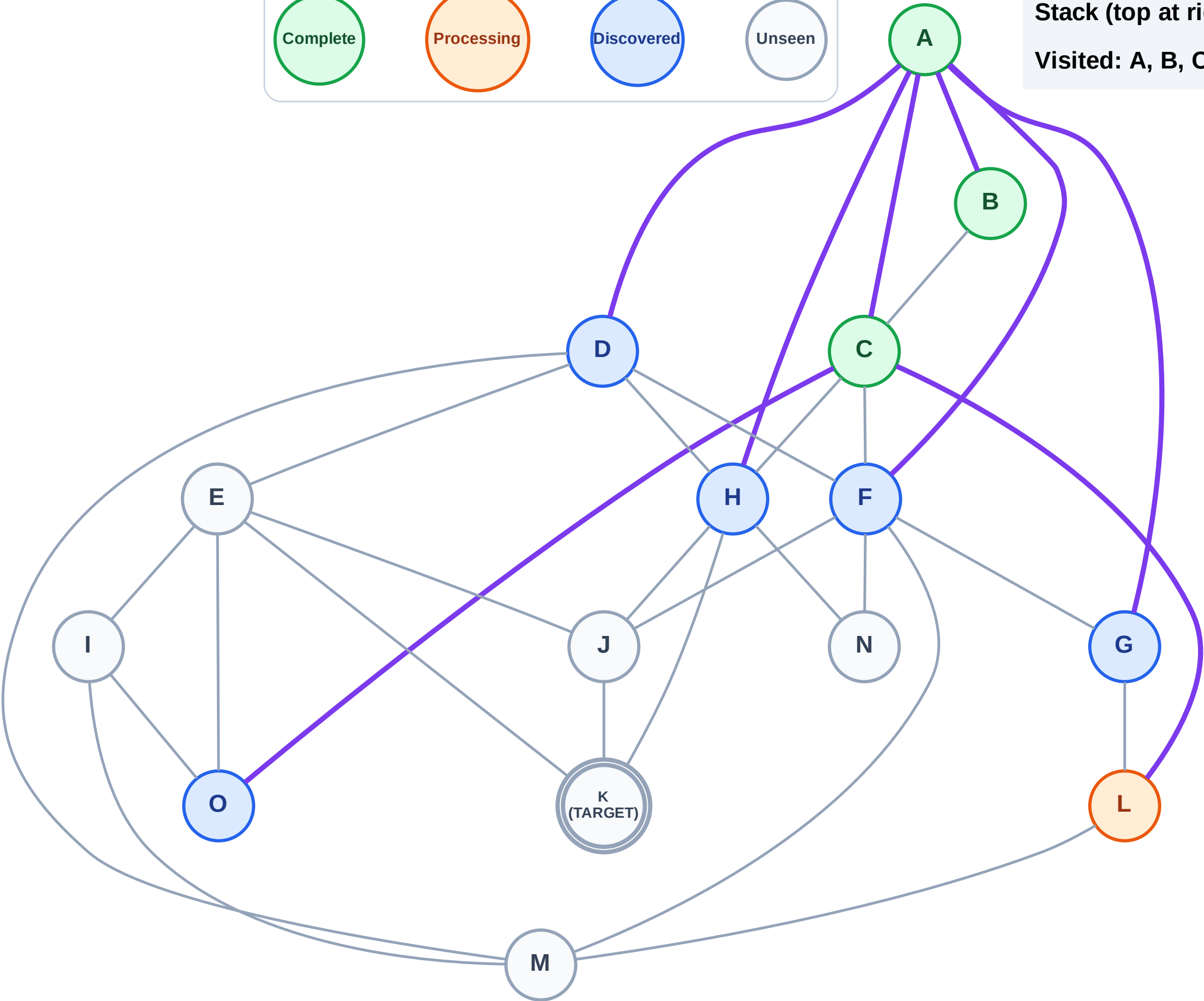
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D | O

Visited: A, B, C



DFS Traversal

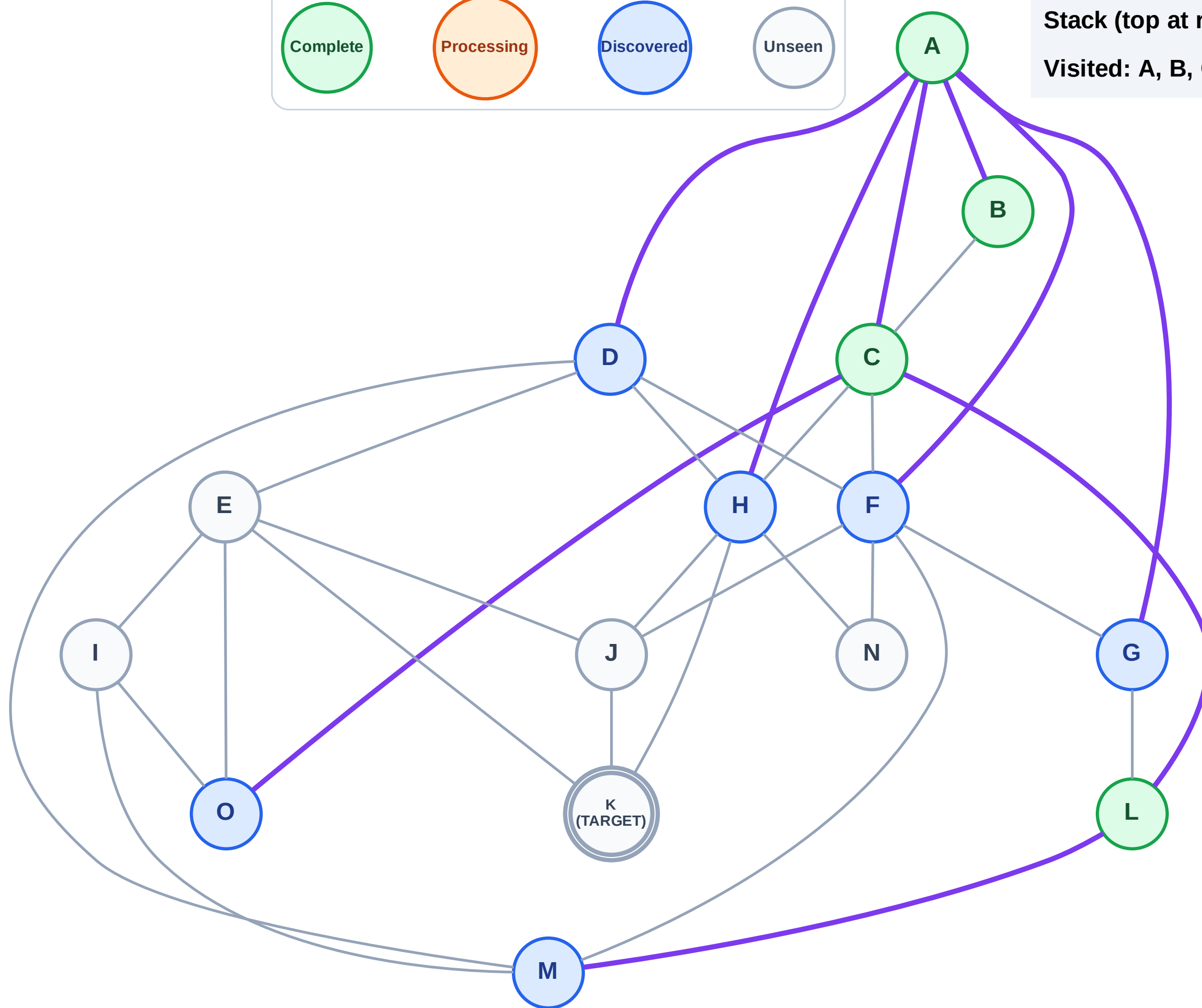
Step 9: Discover M from L

Legend



Stack (top at right): H | G | F | D | O | M

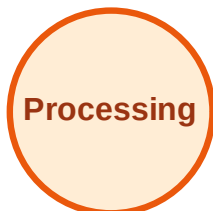
Visited: A, B, C, L



DFS Traversal

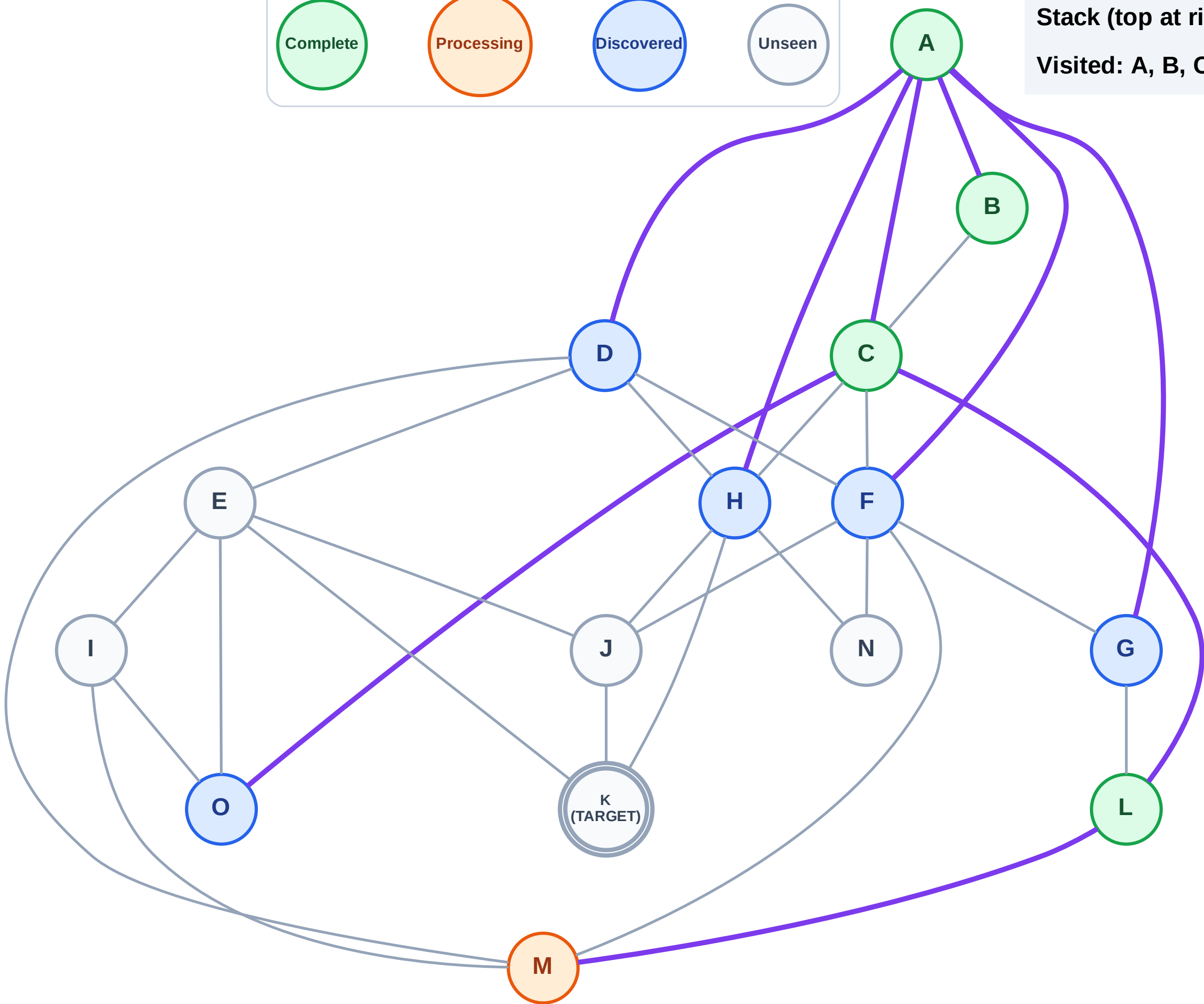
Step 10: Process node M

Legend



Stack (top at right): H | G | F | D | O

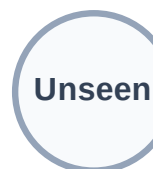
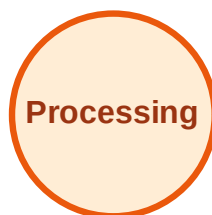
Visited: A, B, C, L



DFS Traversal

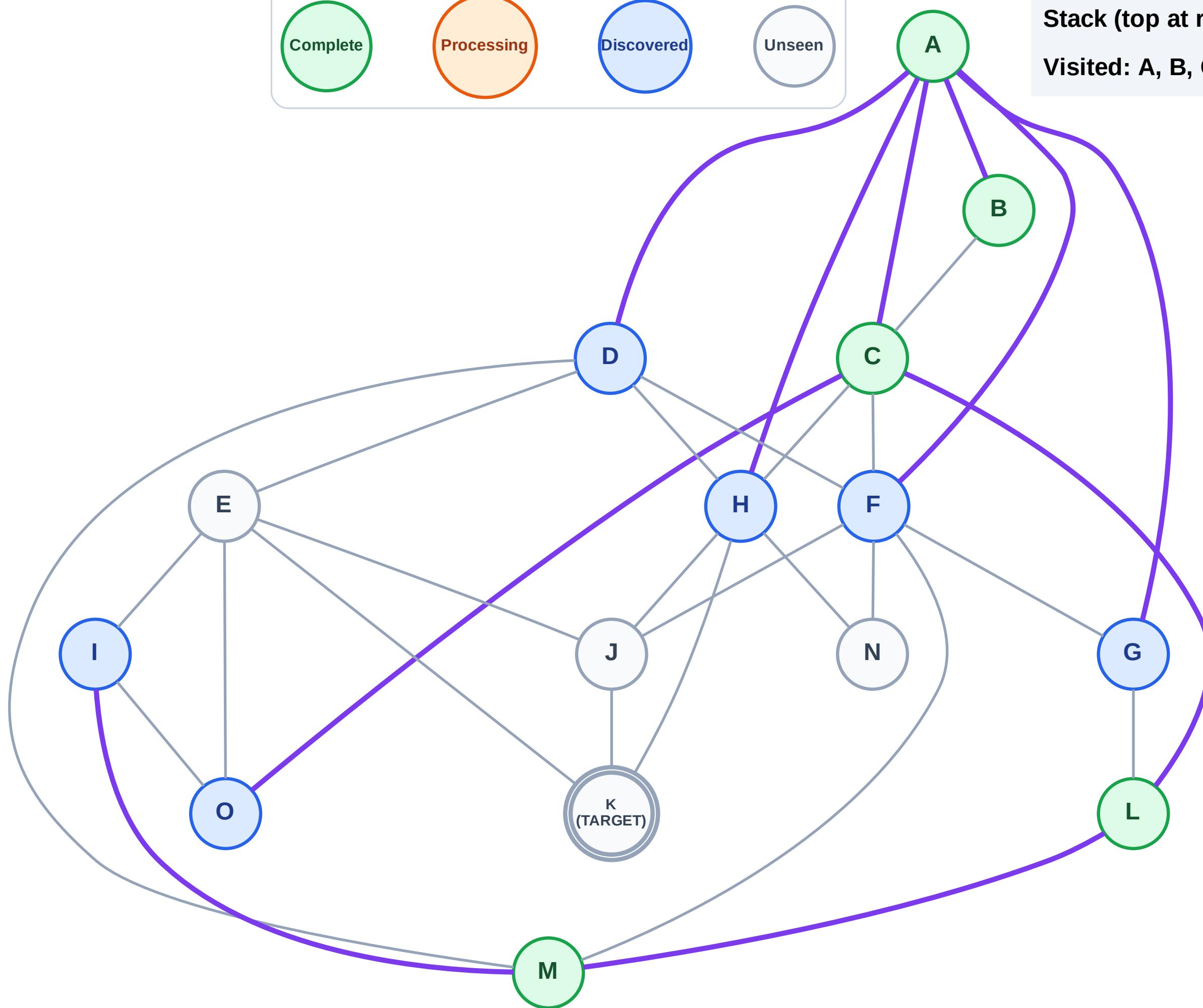
Step 11: Discover I from M

Legend



Stack (top at right): H | G | F | D | O | I

Visited: A, B, C, L, M



DFS Traversal

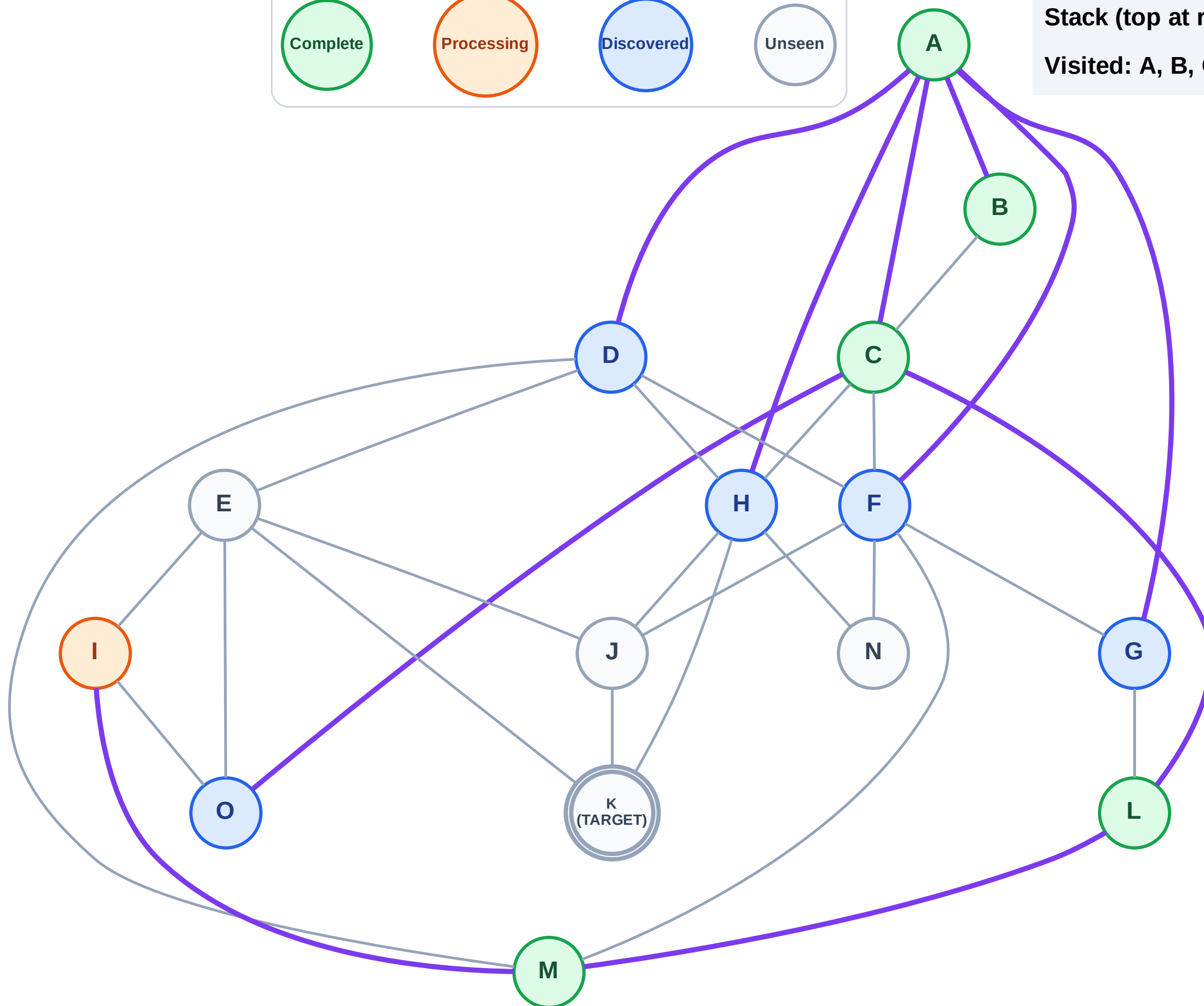
Step 12: Process node I

Legend



Stack (top at right): H | G | F | D | O

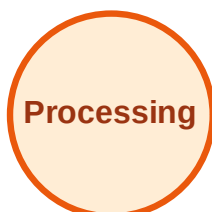
Visited: A, B, C, L, M



DFS Traversal

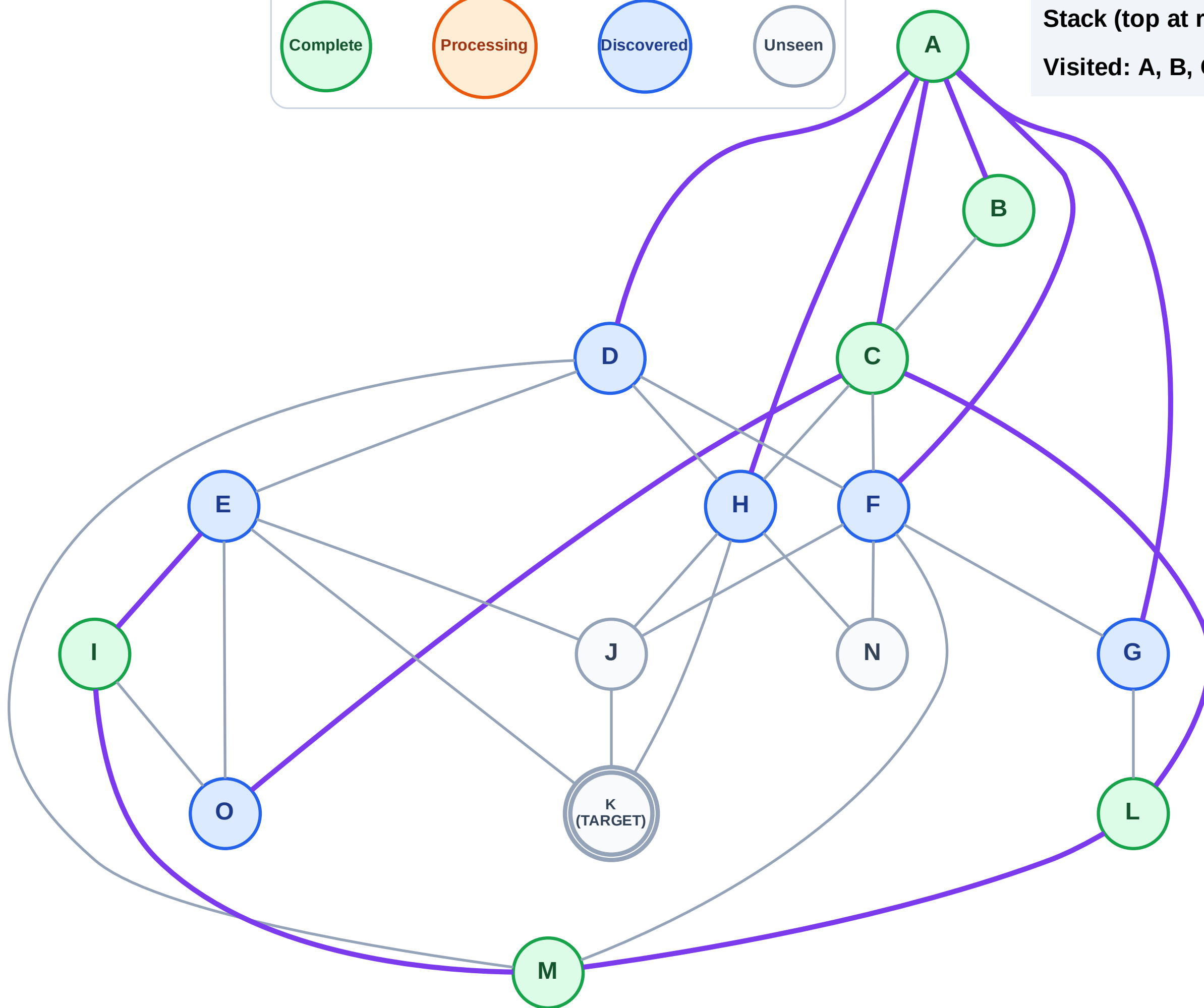
Step 13: Discover E from I

Legend



Stack (top at right): H | G | F | D | O | E

Visited: A, B, C, I, L, M



DFS Traversal

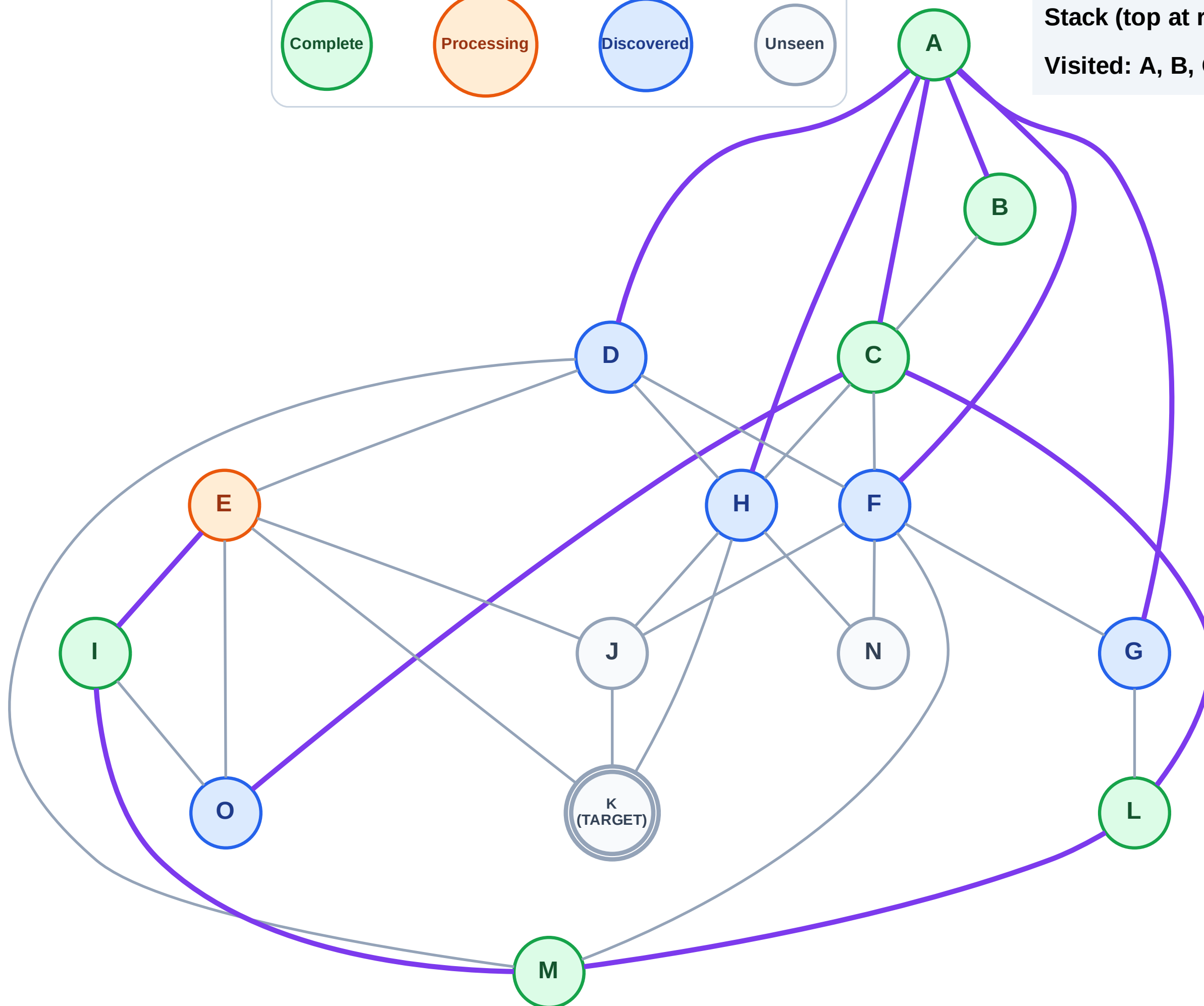
Step 14: Process node E

Legend



Stack (top at right): H | G | F | D | O

Visited: A, B, C, I, L, M



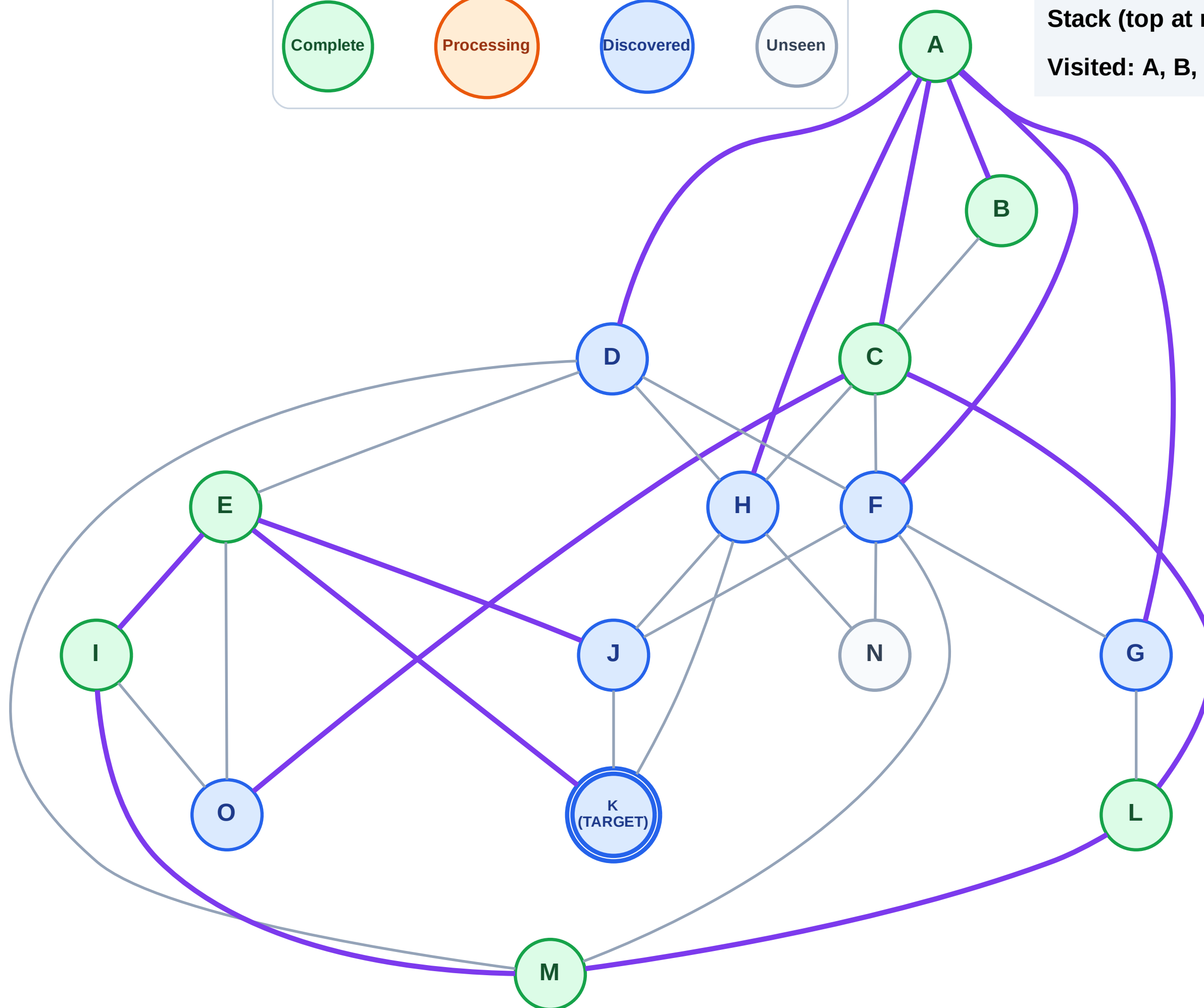
Step 15: Discover K, J from E

Legend



Complete Processing Discovered Unseen

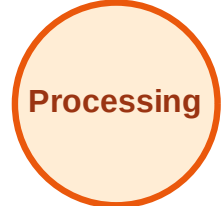
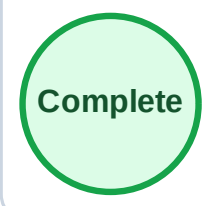
Visited: A, B, C, E, I, L, M



DFS Traversal

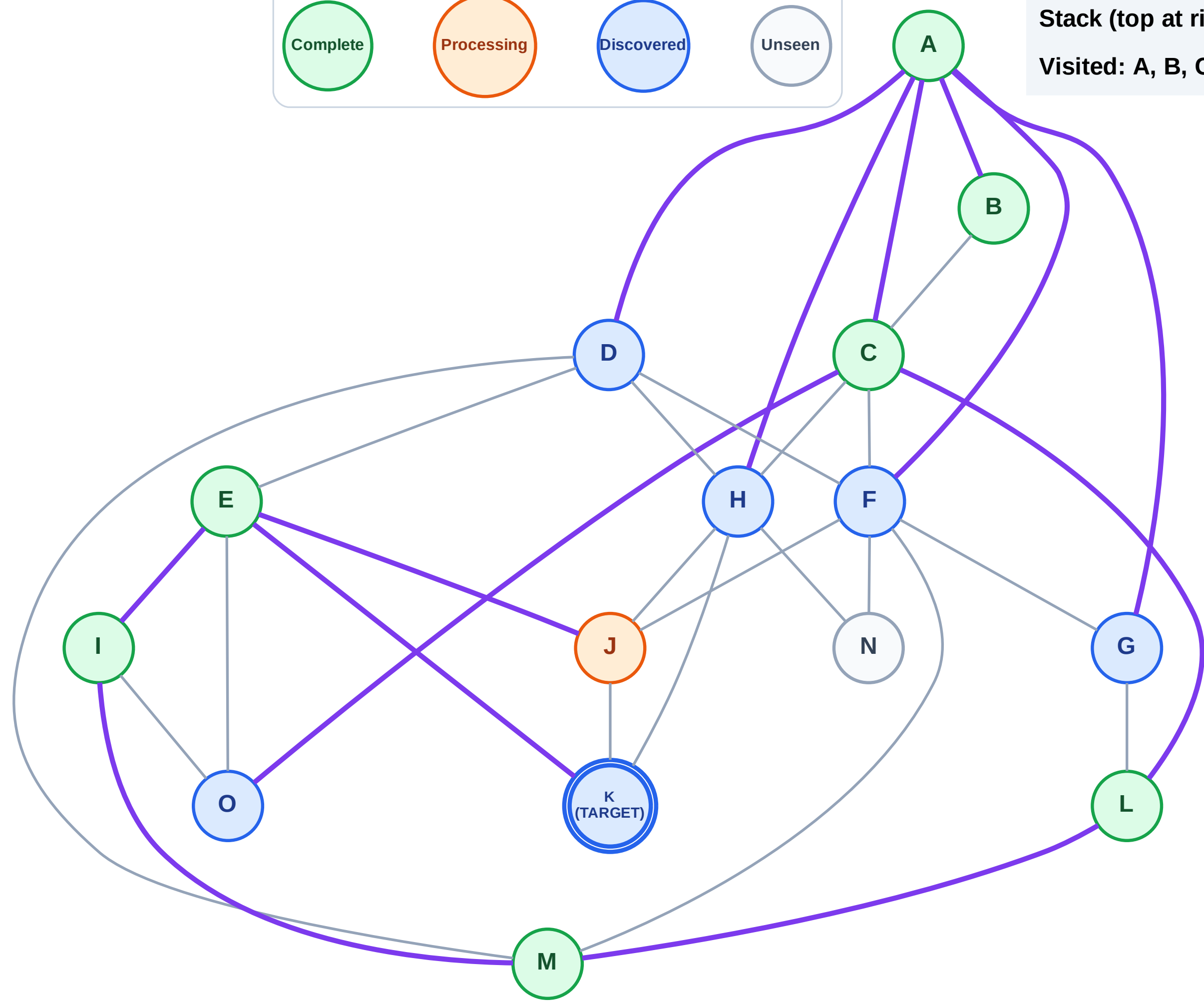
Step 16: Process node J

Legend



Stack (top at right): H | G | F | D | O | K

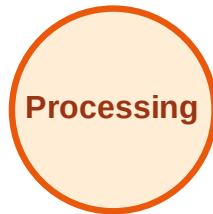
Visited: A, B, C, E, I, L, M



DFS Traversal

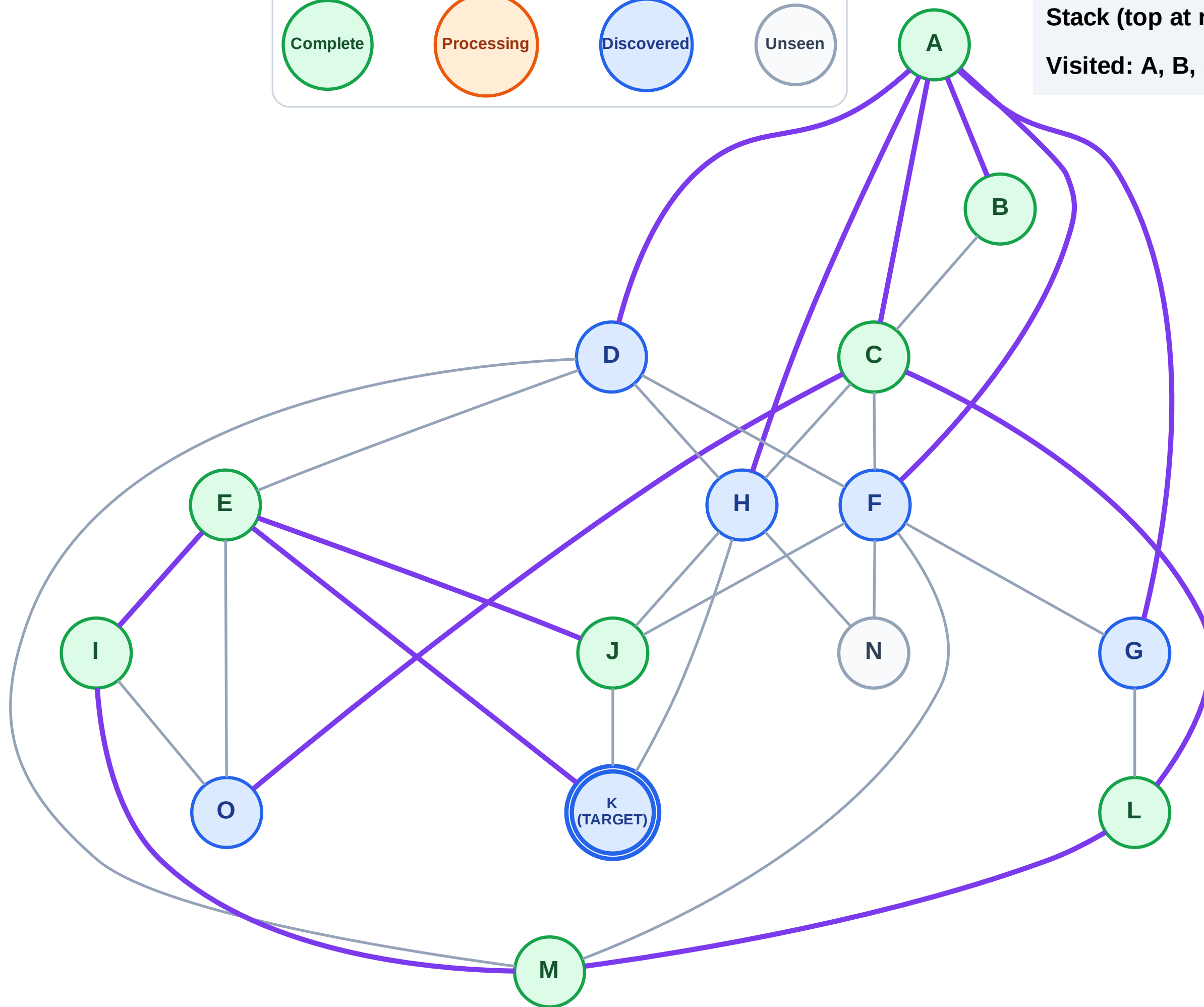
Step 17: No new neighbors from J

Legend



Stack (top at right): H | G | F | D | O | K

Visited: A, B, C, E, I, J, L, M



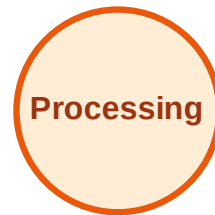
DFS Traversal

Step 18: Process node K

Legend



Complete



Processing



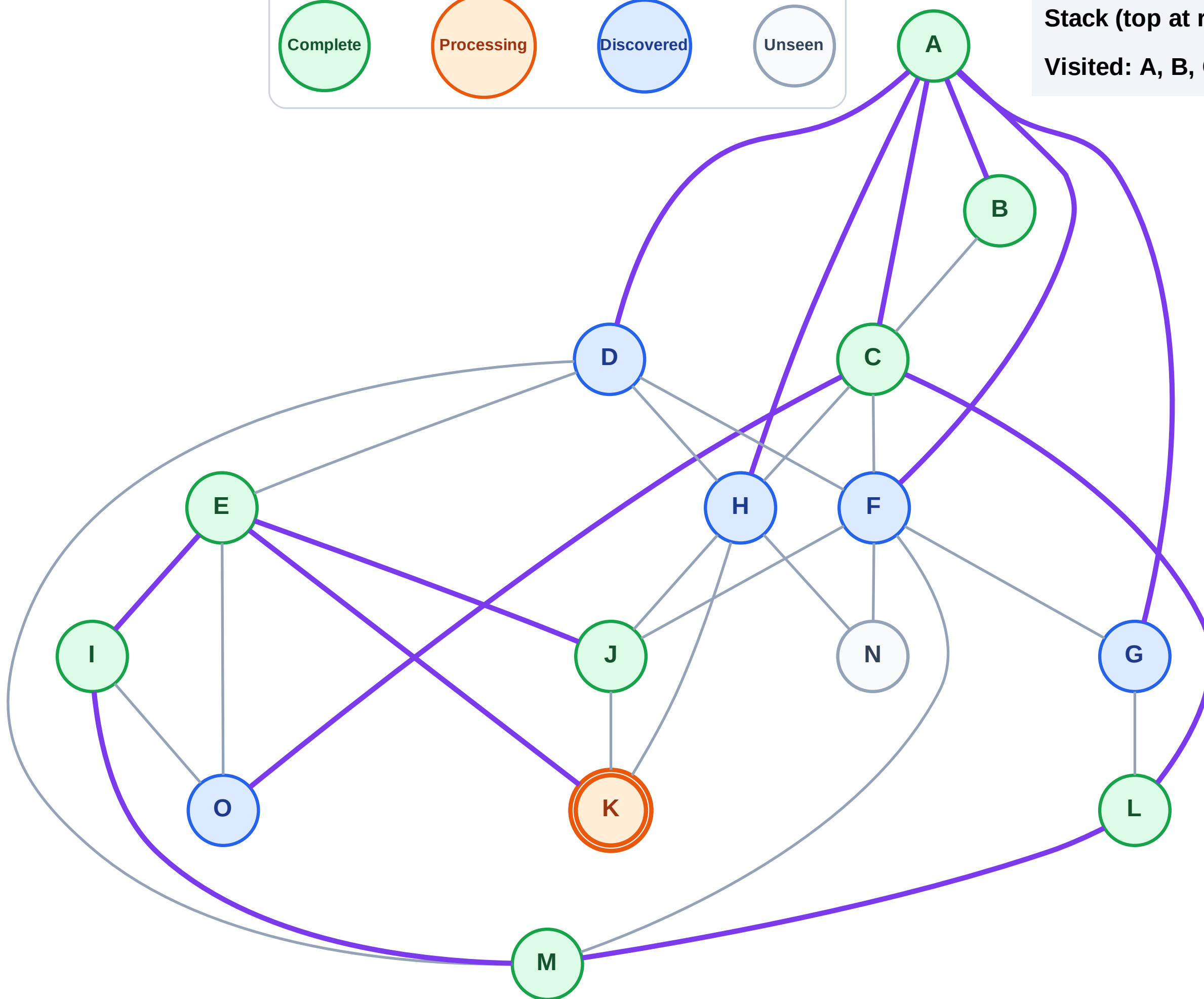
Discovered



Unseen

Stack (top at right): H | G | F | D | O

Visited: A, B, C, E, I, J, L, M



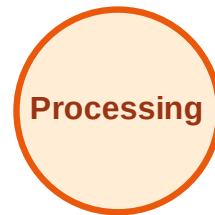
DFS Traversal

Step 19: TARGET NODE K REACHED! Path (6 hops): A → C → L → M → I → E → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): H | G | F | D | O

Visited: A, B, C, E, I, J, K, L, M

